Coastal Erosion Hazard in Bangladesh: Space-time pattern analysis and empirical forecasting, impacts on land use/cover, and human risk perception

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Dissertation submitted to the faculty of the Virginia Polytechnic Institute and State University in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

In

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May 10, 2023

Blacksburg, Virginia

Keywords: Natural hazards, coastal erosion, vulnerability, human displacement, coastal communities, Meghna river, Bangladesh

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ABSTRACT

Coastal areas are vulnerable to different natural hazards, including hurricanes, cyclones, tsunami, floods, coastal erosion, and saltwater intrusion. These hazards cause extensive social, ecological, economic, and human losses. Continued climate change and sea-level rise is expected to substantially impact the people living in coastal areas. Sea level rise poses serious threats for the people living in the coastal zone, which leads to coastal erosion, inundations in the low-lying areas, tidal water encroachment and subsequent salt-water intrusion, as well as the displacement of the people living along the coast.

Coastal erosion is one of the biggest environmental threats in the coastal areas globally. In Bangladesh, coastal erosion is a regularly occurring and major destructive process, impacting both human and ecological systems at sea level. The Lower Meghna estuary, located in southern Bangladesh, is among the most vulnerable landscapes in the world to the impacts of coastal erosion. Erosion causes population displacement, loss of productive land area, loss of infrastructure and communication systems, and, most importantly, household livelihoods. For a lower middle-class country, such as Bangladesh, with limited internal resources, it is hard to cope with catastrophic natural hazards, such as coastal erosion and its related consequences.

This research aims to advance the scientific understanding of past and future coastal erosion risk and associated changes in land change and land cover using geospatial analysis techniques. It also aims to understand the patterns and drivers of human perception of coastal erosion risk. To place the research questions and objectives in content, Chapter 1 includes a brief introduction and literature review of the coastal erosion context in Bangladesh. Chapter 2 assesses different methods of prediction to investigate the performance of future shoreline position predictions by quantifying how prediction performance varies depending on the time depths of input historical shoreline data and the time horizons of predicted shorelines. Chapter 3 evaluates historical land loss and how well predicted shorelines predict amounts of succeeding Land Use

and Land Cover (LULC) resources lost to erosion. Chapter 4 focuses on the patterns and drivers of erosion risk perception using data from spatially explicit measures of coastal erosion risk derived from satellite imagery and a random sample survey of residents living in the coastal communities.

In summary, this research advances our scientific understanding of past and future coastal erosion risk and associated changes in land change and land cover using geospatial analysis techniques. It also enhances the understanding of the patterns and drivers of human perception of coastal erosion risk by combining satellite imagery and social survey data. Compared to much of the coastal erosion literature, this work draws from a 35-year time series of satellite-derived shorelines at annual temporal resolution. This time depth enables us to employ a temporal design strategy expected to yield a robust characterization of space-time erosion patterns. This study also enabled us to assess how well predicted shorelines predict amounts of succeeding LULC resources lost to erosion by using long-term historical data. The innovative we use has potential applications to other deltas and vulnerable shorelines globally. While empirical results are specific to the project's study area, results can inform the region's shoreline forecasting ability and associated mitigation and adaptation strategies.

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GENERAL AUDIENCE ABSTRACT

Coastal erosion is a global problem. Coastal Bangladesh has one of the highest rates of erosion in the world. Erosion causes population displacement, loss of productive land area, loss of infrastructure and communication systems, and, most importantly, household livelihoods. With an aim to advance our understanding of coastal erosion hazard, this study assessed past and future coastal erosion risk and associated changes in land change and land cover and human risk perceptions using different geospatial and statistical analysis techniques. First, different methods of coastal erosion prediction were evaluated to investigate the performance of future shoreline position predictions. Second, the historical land loss was estimated and how well predicted shorelines predict amounts of succeeding land use and land cover (LULC) resources lost to erosion were assessed. Finally, the patterns and drivers of human perception of coastal erosion risk were explored.

DEDICATED TO

My parents

ACKNOWLEDGEMENTS

I am thankful to the following individuals and organizations who have contributed tremendously to this research. It was not possible to complete this dissertation without their assistance.

First, I would like to express my heartfelt gratitude to my advisor, Dr. Thomas W. Crawford. Without my advisor's guidance and support, my dissertation research would not have been so smooth and enjoyable. Thank you so much for encouraging and allowing me to go beyond my dissertation research to develop new skills, network with other researchers and publish with my peers. Your encouragement and support during my five academic years at Virginia Tech helped me to overcome challenges in my research and my daily life. I am also grateful to my committee members Dr. Luke Juran, Dr. Yang Shao, and Dr. Scott Curtis for their service in my committee and valuable advice from the beginning of my dissertation research to my graduation.

I would also like to thank other faculties and staff in the Department of Geography and IGEP Remote Sensing for their assistance and support during my PhD research. I would like to express my gratitude to Dr. James B. Campbell and Dr. Randolph H. Wynne for giving me the opportunity to be part of the IGEP Remote Sensing program. Thanks to Karen Bland, Les Fuller and Stacey Kuhar for their help with everything.

I also want to thank the National Science Foundation, Department of Geography and GEA program, Virginia Tech for providing funding to support my dissertation research. I sincerely appreciate the support from the graduate school at VT, IGEP Remote Sensing and Office of Undergraduate Education. I would also like to thank other collaborators of the NSF project that I have worked on for the last five years. Special thanks to Dr. Munshi Khaledur Rahman for his contribution in this research.

Finally, I want to thank my wife Rasha for her tremendous support and encouragement throughout my PhD journey. I would also like to thank my parents and siblings whose continuous help and support led me to this wonderful success. I am grateful to my Geography and GEA friends specially Ryley Harris, IGEP Remote Sensing friends, SESYNC friends and Bangladeshi friends for being support system throughout my Virginia Tech life.

"Thank you All."

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Chapter 1. Introduction, literature review, and statement of purpose

1.1 Introduction

Coastal erosion is one of the most important phenomena and a major environmental threat in the coastal communities globally (Parry et al. 2007; Malini and Rao 2004). People living in the delta regions are among the worst victims of this problem (Syvitski et al. 2009). In Bangladesh, it is one of the major natural disasters that occurs regularly (Ahmed et al. 2018; Crawford et al. 2020a; Sarwar and Woodroffe 2013). It is well evident that the lower Meghna estuary is one of the most vulnerable areas for coastal erosion. A recent study found that this area has among the highest of erosion rates in the world (Crawford et al., 2020a). Rapid geomorphological changes are taking place in this part of the world due to erosion predominantly; however, new land is also created to accretion of deposited sediments from erosional sites. According to Brammer (2014), the Meghna estuary had a net land gain of 451 km^2 from 1984 to 2007 with an average growth rate of 19.6 km^2 annually. Erosion causes population displacement, loss of productive land area, loss of infrastructure and communication systems, and most importantly household livelihoods (Bhuiyan et al. 2017; Crawford et al. 2020b). Due to extensive erosion in this region, thousands of people are being displaced annually. As a developing country with limited resources, Bangladesh is challenged to cope with the problem of coastal erosion.

1.1.1 Study Area

The study area is located in the eastern bank of Meghna river, near the Bay of Bengal part of one of the biggest deltas in the world. It covers coastal upazilas (local administrative unit) of Chandpur and Lakshmipur districts in Bangladesh. In Bangladesh, the administrative structure consists of Divisions, Districts, Upazilas and Union Parishad. The upazilas are the second lowest tier of administrative unit. The upazilas include Lakshmipur Sadar, Raipur, Ramgati, and

Kamalnagar in Lakshmipur district and Chandpur Sadar and Haim char upazilla in Chandpur district. According to population census of 2011, more than two million people live in the study area's six upazilas. Most households depend on agriculture or fishing for their livelihood (Paul et al. 2020; Rahman et al. 2021). This study covers more than 85 km stretch of the eastern bank of the lower Meghna estuary. According to previous studies, this part of the estuary is highly vulnerable to coastal erosion due to heavy discharge of sediment flow from the upstream of Ganges-Brahmaputra Meghna (GBM) delta, one of the largest deltas in the world. People living in this low-lying coast are very close to the sea level with an average elevation of 5m.

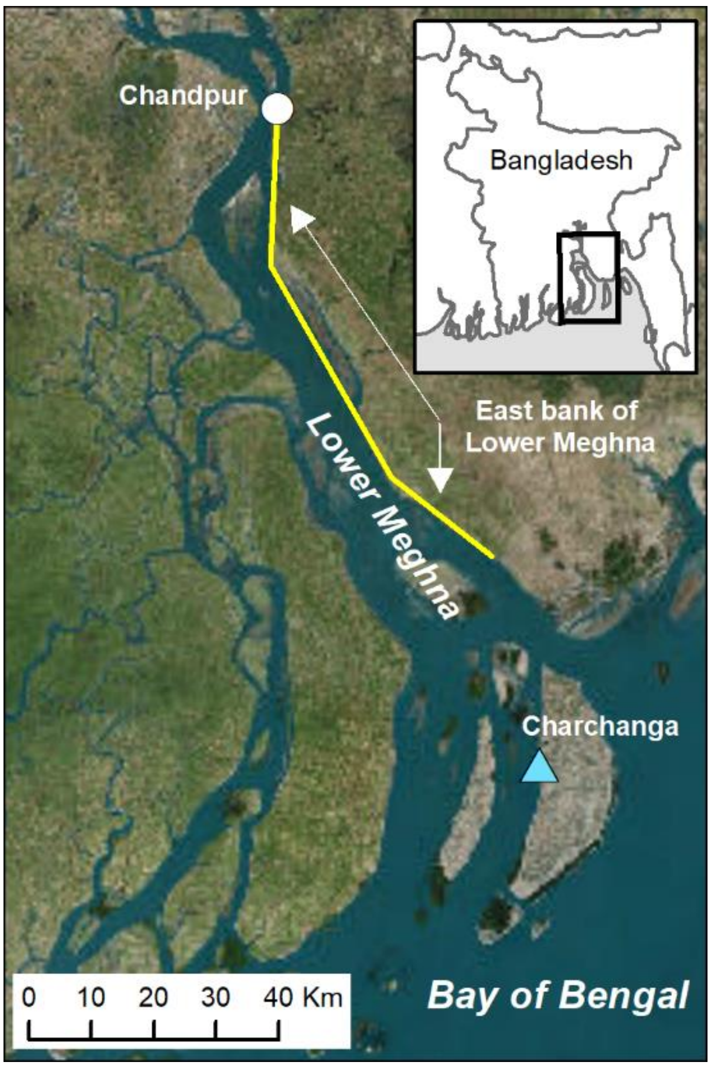

Figure 1. Study area – the eastern bank of lower Meghna River in Bangladesh

1.2 Literature Review

1.2.1 Natural Hazards, Risk and Vulnerability

This work is positioned within the realm of hazards research that is investigated within disciplinary geography and relate disciplines. A natural hazard is a natural phenomenon that might have impacts on humans, other animals and the environment. According to United Nations, natural hazards can be defined as a natural process or phenomenon that may cause loss of life, injury or other health impacts, property damage, loss of livelihoods and services, social and economic disruption, or environmental damage (UNISDR 2009). Natural hazard events occur naturally and can be categorized as geological (coastal erosion, earthquakes, landslides, volcanic eruption), meteorological or climatological (drought, heat wave, cyclone, tornado), hydrological (floods) or biological (disease epidemics). To understand the dynamic relationship between humans and natural hazards, it is critical to understand other concepts like risk, vulnerability, capacity etc.

A risk can be viewed as

$$\text{Risk} = \frac{\text{Hazards} * \text{Vulnerability}}{\text{Capacity to cope}} \qquad (1)$$

The risk is the probability that a hazard will cause deaths or injuries, loss of property or damage to the environment. Vulnerability is an essential concept in hazard research and can be defined as the likelihood that an individual or group will be exposed to and adversely affected by a hazard (Cutter 1993). There are some major factors that influence the vulnerability of the people to a hazard event. These factors include lack of access to resources, lack of political power and representation, beliefs and customs, type and density of infrastructure (Cutter et al. 2003). The coping capacity can be defined as the ability of people, organizations and systems, using available skills and resources, to face and manage adverse conditions, emergencies or disasters (UNISDR

2009). If the risk grows with the increase of global hazards and people's vulnerability, the people's coping capacity decreases.

1.2.2 Risk Perception

Risk perception can be defined as the intuitive of judgement of individuals and groups of risks in the context of limited uncertain information (Slovic 2000). Kellens et al. (2014) defined risk perception as the research that involves people's emotions, awareness, and behavior with regard to hazards. While originating in the 1960s (Starr 1969), the study of risk perception of natural hazards has become more and more important to different hazard studies. Understanding risk perceptions for improving risk communication is important because it may offer important insight of what drives those perception to risk communicators for better managing and planning of natural hazards and disasters.

While drivers of risk perception may vary by system, Carlton and Jacobson (2013) found that there are several factors that are important components of risk perception including social trust, environmental attitudes, risk salience, and demographic factors. Individuals with higher social trust may perceive less risk compared to individuals with less social trust (Bronfman et al. 2016). Risk perceptions are also influenced by environmental attitudes. For instance, Slimak and Dietz (2006) found that higher levels of environmental concerns are related to higher level of risk perception. Another factor that may influence risk perception is risk salience. Generally, risk salience has two components: relevant prior experience and proximity to the risk (Carlton and Jacobson 2013). Previous experience of a natural hazard event might have greater influence on perceived personal risk. A study by Lindell and Hwang (2008) on households' response to multi-hazard environment found that the people with prior exposure to a hazard were more aware than the people without hazard experience. In addition to prior experience, proximity to the risk event has a substantial influence on perceived risk. For example, people living in the coast are more exposed to natural hazard events than the people living inland. Lastly, risk perceptions can be influenced by demographic factors. Many studies have found that factors like gender, political affiliation or educational background influence individuals' perception of risk (Slovic 1999; Linden et al. 2015; Sund et al. 2017). A study on socio-physiological determinants of climate change risk perception by Linden (2015) found that Gender, political party, knowledge of the causes, impacts and responses to climate change, social norms, value orientations, affect and personal experience with extreme weather were all identified as significant predictors. It is also evident that living in a historical hazardous place might also influence risk perception (Lujala et al. 2015).

1.2.3 Coastal hazards

Globally, coastal areas are vulnerable to different natural hazards including floods, cyclones, hurricanes, tsunami, saltwater intrusion, and coastal erosion because of its proximity to the sea. These hazards are responsible for extensive human, financial, social and ecological losses (Sahoo and Bhaskaran 2018). According to UN (2017), about 40% of the world's population live within 100 km (60 miles) of the coast, which is expected to grow fast in the near future. Future climate change and subsequent sea-level rise is expected to impact substantially the vulnerability of the people living in the coastal areas (Klein et al. 2003; Passeri et al. 2015). Sea level rise poses serious threat for the people living in the coastal zone, which leads to coastal erosion, inundations in the low-lying areas, tidal waters encroachment and subsequent salt-water intrusion as well as the displacement of the people living along the coast.

Predicted climate change is adding extra risk for the people living near the coastline. Although the response of world's coastline will vary by region, it is predicted that the poor countries will suffer more regarding future climate change. A study on the distributional impact of climate change on rich and poor countries by Mendelsohn et al. (2006) found that although wealth, adaptation and technology may influence distributional consequences across countries, the major reason why the poor countries are so vulnerable is their location. People living in the same communities are also disproportionately impacted by the fate of climate change. Among the same population group, poor people may be heavily impacted by climate change even when impacts on the rest of people is limited. A report from the Demographic and Health Survey (DHS) conducted by the World Bank in multiple countries suggest that the poor people are losing two to three times more than non-poor people when hit by any natural disaster (Park et al. 2015; Hallegatte and Rozenberg 2017).

1.2.4 Coastal Erosion

Coastal erosion is one of the most prominent problems in the coastal areas globally. and the world's mega deltas are heavily affected (Woodroffe et al. 2006). According to IPCC WGI (2001), sea level has risen 10–20 cm during the 20th century, possibly exacerbating coastal erosion globally. Due to rising seas, the resulting inundation will heavily impact low-lying areas; impacting at least 100 million persons globally who live within one meter of mean sea level (Zhang et al., 2004). This is a tremendous threat for the people living in the coast as most part of the coastal areas in Bangladesh are within few meters from the sea level.

Coastal communities in Bangladesh are vulnerable to different natural hazards and disasters. Extreme erosion is one of the biggest problems that the coastal people are facing. Coastal Bangladesh is highly vulnerable to coastal erosion, where land erosion and accretion are among the highest in the world (Brammer 2014; Ahmed et al. 2018; Crawford et al. 2020a). It is a recurring problem causing thousands of people homeless every year. The poor people living in the coastal areas are the worst victim of such changes. Bangladesh is considered as one of the most vulnerable countries in the world to the impacts of climate change, particularly sea-level rise (Nicholls et al. 2007; Sarwar and Khan 2007). Moreover, future global warming and subsequent sea level rising possess serious threat to the people living in this part of the world. Additionally, as economically marginalized populations live in coastal Bangladesh, the magnitudes and severities of the affected people may affect the total economic growth of the country (Hossain et al. 2016).

Due to coastal erosion, people living on the coast not only lose their agricultural land but also infrastructure, communication systems, and most importantly their livelihood. As a lower middle-class country like Bangladesh, with limited internal resources, it is hard to cope with catastrophic natural hazards like erosion and its related consequences (Poncelet et al. 2010). Additionally, disasters like coastal erosion create the acute problems of unemployment in coastal rural areas and thereby, worsen the socio-economic condition of the displaced people. Most of these displaced people become landless, which pushes them into further poverty and forced migration. Future climate change is more likely to impact the migration decisions of the people living in coastal Bangladesh in search of livelihoods (Call et al. 2017).

1.2.5 Estimation of Shoreline Change Rates

Remote sensing and GIS techniques are useful and widely used by coastal researchers in detecting coastal erosion and accretion (Sarwar and Woodroffe 2013; Kaliraj et al 2017; Ahmed et al. 2018). In quantifying shoreline movement, different methods are used including the End Point Rate (EPR), Net Shoreline Movement (NSM), Linear Regression Rate (LRR), and Weighted Linear Regression (WLR) (Sarwar and Woodroffe 2013; Mullick et al. 2019; Jana 2020).

The Digital Shoreline Analysis System (DSAS) is the most used toolkit in detecting shoreline movement. The DSAS toolkit is an ArcGIS plugin developed by the United States Geological Survey (Thieler et al. 2009). It consists of digital inputs and outputs as implemented within the DSAS software and defined by the user: 1. A digital baseline GIS polyline feature(s), 2. A set of digital transects that are cast orthogonal to the baseline, 3. A set of two or more vector shorelines for different dates, and 3. output measures of shoreline change rates (erosion or accretion per annum) specific to each transect.

Among the methods used, the most widely accepted methods are the End Point Rate (EPR), Linear Regression Rate (LRR), and Weighted Linear Regression (WLR). The End Point Rate is estimated by dividing the distance of shoreline movement along the transect by the time elapsed between the oldest and the most recent shoreline. The Linear Regression Rate (LRR) is estimated by fitting a least-squares regression line to all shoreline points for a particular transect. Positive EPR and LRR values represent the rate of shoreline erosions (m/y), and negative values represent the rate of erosion (m/yr). Lastly, the Weighted Linear Regression can be defined as the generalization of linear regression in which uncertainty of the measurement is incorporated into the regression.

1.2.6 Prediction of Future Shoreline Positions

Estimation of shoreline change rates is based on prior observations of historical shoreline. Prediction of future shorelines can incorporate historical rates and/or other data and information. Future shoreline prediction is important for informed management of the coastal zone and to limit the structural and financial losses in the coast. Due to the dynamic nature of physical coastal processes, prediction of future shorelines can be particularly challenging. Shoreline retreat (erosion) and advance (accretion) can depend on many factors and processes including fluvial discharge, bathymetry, land cover along the littoral, wind and wave energy, rainfall, sea level rise, human alterations (e.g., protective structures), and acute disturbance events such as cyclones.

There are two general approaches to shoreline prediction. First, researchers have used physically based process models that involve various geophysical input parameters (e.g., discharge, wave energy, etc.) (Ibaceta et al. 2020; Vitousek et al. 2017). Coastal geoscientists and engineers often apply this approach. A second approach employs empirically based, data-driven statistical methods that do not involve process models or simulations. This type of prediction uses historical observations of erosion/accretion rates to predict future scenarios. Geospatial scientists commonly use this approach.

This research adopts the second approach and is an empirically based, data driven approach of shoreline prediction. Historical change rates have been often been used to predict future shoreline by employing statistical extrapolation (Mondal et al. 2020; Patel et al. 2021) or machine learning methods. Key inputs are historical change rates. As described previously, different methods yield different rates that can be alternatively employed; e.g., End Point Rate (EPR), Linear Regression Rate (LRR), and Weighted Linear Regression rate (WLR). Prior research has used different methods including EPR (Mukhopadhyay et al. 2012; Mondal et al. 2020), Kalman filter (Ciritci and Turk 2020), Linear Regression Rate (LRR) (Patel et al. 2021), and Artificial Neural Network (ANN) (Zeinali et al. 2021). In recent studies, artificial neural networks and other advanced machine learning techniques have been used prediction of the shoreline movement (Peponi et al., 2019; Kumar et al., 2020).

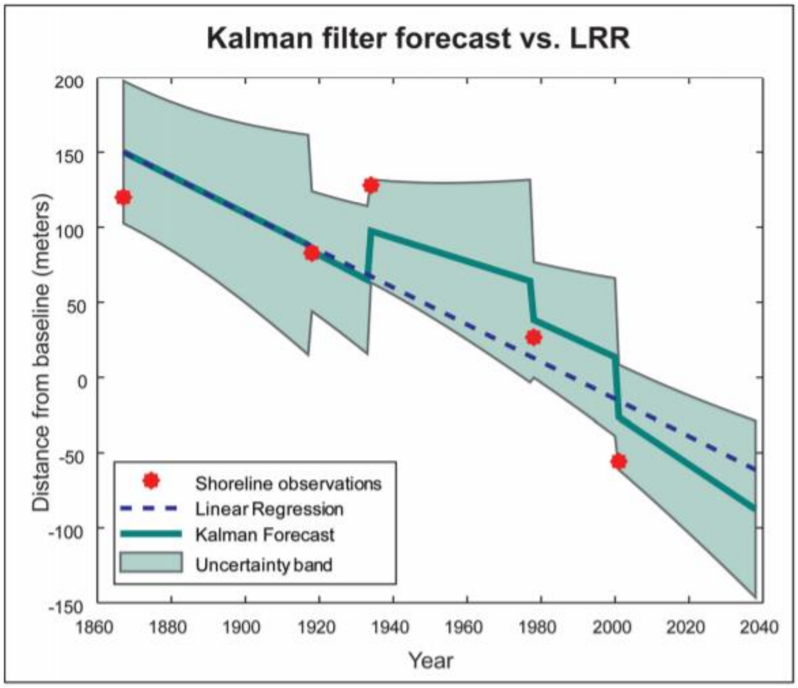

Figure 2. A comparison of the shoreline forecast using Kalman filter versus linear regression (Himmelstoss et al. 2018)

Among the methods used, the End Point Rate (EPR) is one of the most widely used method due to its easy to use and no requirement of additional data or use of a physically based model (Al-Zubieri et al., 2020). It is argued that Kalman filter method is better than the extrapolation of EPR or LRR. The main difference between Kalman filtering method and normal extrapolation is that Kalman filter method is a recursive algorithm that employs multiple steps applied to estimates of the current state variable (e.g., EPR) at each successive time step along with its uncertainty. The outcome of successive estimates of future shoreline movement, which includes spatial uncertainty and random noise, is updated using a weighted average with more weight being given to estimates with higher certainty. A recent study by Ciritci and Turk (2020) used the Kalman filter method to compare the prediction accuracy alternatively using EPR, LRR, and WLR as inputs using a future prediction time horizon of 10 and 20 years. Results suggested that using the WLR shoreline change rate as input to the Kalman filter prediction method provided more accurate results than with use of the other rates (e.g., EPR – End Point Rater; LRR – Linear Regression Rate).

1.2.7 Land Use and Land Cover Change (LULCC) in the coast

A proper estimation of land use and land cover change is crucial for better understanding of the natural and anthropogenic changes influencing the environment. It is particularly important for proper planning and implementation of appropriate environmental practices. Land use land cover change using satellite imagery is a cost-effective way to monitor long-term changes. It is well evident that in the recent decades, human activity has significantly influenced land use and land cover globally. A significant portion of the global population lives near the coast. As a result, it is expected that the coastal ecosystem is going to be changing globally due to a large number of human footprints in the coastal region. For regions experiencing the loss or gain of land resources due to erosion/accretion processes, it is important to map existing land use and cover (LULC) and to integrate resulting data products with information on future shoreline change.

This research apply competing LULC classification algorithms to classify coastal LULC. Recent literature on land use/cover classification in the coastal Bangladesh reveals that the Maximum Likelihood algorithm has been most commonly applied for this task. Table 1 reports classification methods used by recent research of coastal Bangladesh. These algorithms are not the only approaches to LULC classification. Other work has used methods such as support vector machine (SVM), Random Forests (RF), classification and regression trees (CART), and other algorithms. This work apply two different LULC algorithms to assess optimal classification performance. I use the Random Forest (RF) and Support Vector Machine (SVM) algorithms. Following the classifications schemes common to published literature on coastal Bangladesh, I use a target LULC classification scheme with the output categories agriculture, built-up/developed, water body, bare land, and forest.

Table 1 – Selected articles on LULC classification in the coastal Bangladesh

Source	Classification Algorithm
Abdullah et al. 2019	Random Forest (RF)
Hoque et al. 2020	Maximum likelihood
Islam et al. 2020	Maximum likelihood
Rahman et al. 2017	Maximum likelihood
Islam et al. 2021	Maximum likelihood
Ahmed et al. 2021	Maximum likelihood
Chowdhury et al. 2020	Maximum likelihood
Pal and Talukdar 2020	Maximum likelihood

A recent study by Abdullah et al. (2019) found that the vegetation coverage in the coastal Bangladesh is declining day by day due to natural (e.g. flooding, cyclones etc.) as well as human influenced alterations (e.g. shrimp farming). They also found that river areas have been increased by 4.52 percent within 28 years (1990 – 2017). This indicate that as this part of the country is the heart of the Ganges delta, coastal areas of Bangladesh is experiencing erosion at an alarming rate over the last few decades. A similar study by Rahman et al. (2017) found an increase of 26% in shrimp farm area from 1989 – 2015. During this 26 year of period, about 21% of bare lands declined. This is an indication that this highly dynamic coastal region of Bangladesh is being changed rapidly.

1.3 Statement of purpose, significance, and dissertation outline

1.3.1 Statement of purpose

This research aims to advance the scientific understanding of past and future coastal erosion risk and associated changes in land change and land cover using geospatial analysis techniques. It also aims to understand the patterns and drivers of human perception of coastal erosion risk. This research introduces social dimensions to the analysis of erosion hazard by investigating the drivers of human perception of erosion risk. In this project, I quantify past coastal erosion rates and empirically predict future shoreline positions. One goal is to assess the variability of future shoreline prediction performance depending on the time depth of input predictor data (e.g., number of annual historical shoreline observations) and the forecast horizon of predicted future shorelines (e.g., the number of years in to the future that are predicted). I also produce a time-series classification of historical land use and land cover (LULC) change to both quantify LULC change and to evaluate the utility of the future shoreline predictions for predicting amounts of lost or newly added land resources by LULC type.

Last, I use spatially explicit measures of coastal erosion risk (e.g. erosion rates derived geospatially) and human perceptions of erosion risk to investigate risk perception and the degree to which it agrees with the geospatially-derived measures of erosion risk. This focus on risk perception is designed to explain how socio-economic characteristics and locational attributes explain variability in risk perception measures obtained from a household survey conducted in 2018 (n=420).

1.3.2 Research Significance

This research contributes to various subfields of geographical research and related disciplines. It is situated broadly within the context of natural hazards and engages at more detail with applied geospatial analysis of coastal environments, land use and cover (LULC), and risk perception.

First, compared to much of the coastal erosion literature, this work draws from a 35-year time series of satellite-derived shorelines at annual temporal resolution. This time depth enables me to employ a temporal design strategy expected to yield a robust characterization of space-time erosion patterns. It also enables me to empirically assess the performance of future shoreline position predictions by quantifying how prediction performance varies depending on the time depths of input historical shoreline data and the time horizons of predicted shorelines. This is a methodological innovation with potential for forecasting applications to other deltas and vulnerable shorelines globally. While empirical results are specific to the project's study area, results can inform the region's shoreline forecasting ability and associated mitigation and adaptation strategies.

Second and turning to the project's social science dimensions, this work uses household survey data and seeks to understand the patterns and drivers of risk perception reported in survey data and actual risk, with actual risk being observed by empirical geospatial methods and products. A regression model was designed to understand how socio-economic and locational attributes explain the variability of respondents' accuracy in risk perception. This approach adds to the hazards and risk perception literature, which typically does not investigate the drivers of risk perception combining both survey data and satellite imagery.

1.3.3 Research Questions

1. How does coastal erosion/accretion vary spatially and temporally?
2. How successful is empirical forecasting of future shoreline positions?
    - How well do alternative extrapolation methods of shoreline prediction perform?
    - What time depths of historical predictor shorelines and what time horizons of future predicted shorelines yield the best predictive performance?

- How does predictive performance vary over space and time?

3. What are the impacts of shoreline change on land use and land cover (LULC)?
    - How much land has been lost in the lower Meghna estuary region?
    - How much LULC has been lost to erosion over defined time intervals?
    - How well do predicted shorelines predict amounts of succeeding LULC resources lost to erosion?
4. What socio-economic and locational attributes explain variation in individuals' risk perception?
    - What is the relative importance of age, gender, income, length in residence, prior erosion displacement experience, location, and other factors in explaining risk perception of coastal erosion?

1.3.4 Structure of the dissertation

This dissertation is organized in five chapters. Chapter 1 provides a brief introduction and literature review relevant to natural hazards, coastal erosion, risk perception, erosion prediction, and LULC. Chapter 2 focuses on different methods of prediction to investigate the performance of future shoreline position predictions. Chapter 3 estimates historical land loss and evaluates how well predicted shorelines predict amounts of succeeding LULC resources lost to erosion. Chapter 4 assesses the patterns and drivers of erosion risk perception of the people living in the coastal communities. Chapter 5 provides overall conclusions of the study.

1.4 Data sources

This study utilized data from different sources. The datasets used in this study are outlined in Table 2.

Table 2. Datasets used to conduct this study

Data	Source
Landsat Imagery	USGS Earth Explorer
Tidal Data	Bangladesh Water Development Board
Population Data	Bangladesh Bureau of Statistics
Satellite Imagery	Google Earth Pro
Socio-economic, demographic, and risk perceptions data	Social Survey

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Chapter 2. Assessment of spatio-temporal empirical forecasting performance of future shoreline positions

This chapter analyzes spatio-temporal empirical forecasting performance of future shoreline positions. The manuscript from this chapter was published in the journal "*Remote Sensing*".

Islam, M. S., & Crawford, T. W. (2022). Assessment of Spatio-Temporal Empirical Forecasting Performance of Future Shoreline Positions. *Remote Sensing*, *14*(24), 6364.

Abstract

Coasts and coastlines in many parts of the world are highly dynamic in nature, where large changes in the shoreline position can occur due to natural and anthropogenic influences. The prediction of future shoreline positions is of great importance in the better planning and management of coastal areas. With an aim to assess the different methods of prediction, this study investigates the performance of future shoreline position predictions by quantifying how prediction performance varies depending on the time depths of input historical shoreline data and the time horizons of predicted shorelines. Multi-temporal Landsat imagery, from 1988 to 2021, was used to quantify the rates of shoreline movement for different time period. Predictions using the simple extrapolation of the end point rate (EPR), linear regression rate (LRR), weighted linear regression rate (WLR), and the Kalman filter method were used to predict future shoreline positions. Root mean square error (RMSE) was used to assess prediction accuracies. For time depth, our results revealed that the higher the number of shorelines used in calculating and predicting shoreline change rates the better predictive performance was yielded. For the time horizon, prediction accuracies were substantially higher for the immediate future years (138 m/year) compared to the more distant future (152 m/year). Our results also demonstrated that the forecast performance varied temporally and spatially by time period and region. Though the study area is located in coastal Bangladesh, this study has the potential for forecasting applications to other deltas and vulnerable shorelines globally.

Keywords: coastal erosion; vulnerability; digital shoreline analysis system; shoreline forecasting; delta; Bangladesh

Introduction

Globally, coastal areas are vulnerable to different natural hazards, including floods, erosion, hurricanes, cyclones, tsunami, and saltwater intrusion, because of their proximity to the sea. These hazards are responsible for extensive economic, social, ecological, and human losses [1]. According to the United Nations [2], approximately 40% of the world's population lives within 60 miles of the coast, which is expected to grow significantly in the future. Continued climate change and sea-level rise is expected to substantially impact the people living in coastal areas [3,4]. Sea level rise poses serious threats for the people living in the coastal zone, which leads to coastal erosion, inundations in the low-lying areas, tidal water encroachment and subsequent salt-water intrusion, as well as the displacement of the people living along the coast.

Coastal erosion is one of the biggest problems in the coastal areas globally, and the world's mega deltas are heavily affected [5]. During the twentieth century, the mean sea level rose by 11–16 cm [6,7], possibly exacerbating coastal erosion globally. Due to rising seas, the resulting inundation will heavily impact low-lying areas; impacting at least 100 million persons globally who live within one meter of the mean sea level [8]. Coastal Bangladesh is highly vulnerable to coastal erosion, where land accretion and erosion are among the highest in the world [9–12]. It is a recurring problem, causing thousands of people to become homeless every year [13]. Economically challenged populations living in coastal areas are among the most severely impacted victims of such changes. Additionally, the magnitudes and severities of the affected people and their associated livelihoods may affect the total economic growth of the country [14]. Due to coastal erosion, people living on the coast not only lose their agricultural land but also their infrastructure, communication systems, and most importantly their livelihoods. For a lower middle-class country, such as Bangladesh, with limited internal resources, it is hard to cope with catastrophic natural hazards, such as coastal erosion and its related consequences [15].

The detection of shoreline movement and prediction of future shoreline position is important for the informed management of the coastal zone and to limit the structural and financial losses in the coast [16]. Due to the dynamic nature of physical coastal processes, the prediction of future shorelines can be particularly challenging. Shoreline retreat (erosion) and advance (accretion) can depend on many factors and processes, including fluvial discharge, bathymetry, land cover along the littoral, wind and wave energy, rainfall, sea level rise, human alterations (e.g., protective structures), and acute disturbance events such as cyclones [17,18]. There are two general approaches to shoreline prediction. First, researchers have used physically based process models that involve various geophysical input parameters (e.g., discharge, wave energy) [19,20]. Coastal geoscientists and engineers often apply this approach. A second approach employs empirically based, data-driven statistical methods that do not involve process models or simulations. This type of prediction uses the historical observations of erosion/accretion rates to predict future scenarios [21,22]. Geospatial scientists commonly use this approach.

Remote sensing and GIS-based techniques are useful and widely used by coastal researchers in detecting and predicting coastal erosion and accretion [10,23,24]. The Digital Shoreline Analysis System (DSAS) is one of the most used toolkits in detecting shoreline movement. The DSAS toolkit is an ArcGIS plugin developed by the United States Geological Survey [25]. In this research, we adopted an empirically based, data-driven approach of shoreline prediction using DSAS to derive rates of shoreline positional change. Historical change rates have been often been used to predict future shorelines by employing statistical extrapolation [26,27]. In this approach, the key inputs are historical change rates derived from satellite imagery.

Among the methods used, the end point rate (EPR) is one of the most widely used method due to its relative ease to use employing two mapped shorelines for two dates and no requirement of additional data or use of a physically-based model [28]. The linear regression rate (LRR) and weight linear regression rate (WLR) described below employ multiple shoreline dates with the associated regression techniques. To obtain the predicted future shorelines, simple extrapolation extends a terminal date shoreline to some date in the future via the simple extrapolation of the observed historical rates of shoreline positional change. The Kalman filter method is an alternative approach for future shoreline prediction. The main difference between the Kalman filter method and simple extrapolation is that the Kalman filter method is a recursive algorithm that employs multiple steps applied to estimates of the current state variable (e.g., EPR) at each successive time step, along with its uncertainty [29]. The outcome of successive estimates of future shoreline movement, which includes spatial uncertainty and random noise, is updated using a weighted average with more weight being given to estimates with a higher certainty. A recent study by Ciritci and Turk [21] used the Kalman filter method to compare the prediction accuracy alternatively using the end point rate (EPR), linear regression rate (LRR), and weighted linear regression (WLR) as inputs using a future prediction time horizon of 10 and 20 years. Results suggested that using the WLR shoreline change rate as input to the Kalman filter prediction method provided more accurate results than with the use of the other rates (e.g., EPR, LRR).

The main goal of this research is to assess the variability of future shoreline prediction performance depending on the time depth of input predictor data (e.g., number of annual historical shoreline observations) and the forecast horizon of predicted future shorelines (e.g., the number of years into the future that are predicted). Compared to much of the coastal erosion literature, this work uses a richer time series dataset derived from a 34-year time series of satellite-derived shorelines at annual temporal resolution. This time depth enables us to employ a temporal strategy to yield a robust characterization of the space-time patterns of shoreline change and future shoreline prediction. It enables us to empirically assess the performance of the future shoreline position predictions by quantifying how the prediction performance varies depending on the time depths of the input historical shoreline data and the time horizons of predicted shorelines. This is a methodological innovation with the potential for forecasting applications to other deltas and vulnerable shorelines globally. While empirical results are specific to the project's study area, results can inform the region's shoreline forecasting ability and associated mitigation and adaptation strategies.

The specific objectives of this study are to (1) assess how coastal erosion/accretion varies spatially and temporally, (2) measure how predictive performance varies over space and time, (3) examine how successful the empirical forecasting of future shoreline positions is, (4) evaluate how well alternative extrapolation methods of shoreline prediction perform, and (5) identify what time depths of historical predictor shorelines and what time horizons of future predicted shorelines yield the best predictive performance.

2 Materials and Methods

2.1. Study Area

The study area is located along the eastern bank of the lower Meghna River, near the Bay of Bengal (Figure 1). This part of the delta is morphologically very dynamic in nature. The Meghna River is an alluvial meandering river which confluences with the Padma River at the upstream of our study area. Its shoreline spans the coastal upazilas (local administrative regions equivalent to US counties) of Chandpur and Lakshmipur, districts of Bangladesh. The upazilas are the second lowest tier of administrative unit. According to the population census of 2011, more than two million people live in the study area's six upazilas. Most households depend on agriculture or fishing for their livelihoods [30,31]. This study covers more than 85 km of shoreline along the eastern bank of the lower Meghna Estuary. According to previous studies [11,12], this part of the estuary is highly vulnerable to coastal erosion due to the heavy discharge of sediment flow from the upstream of one of the largest deltas in the world, named the Ganges–Brahmaputra–Meghna (GBM) delta. People living in this low-lying coast are very close to the sea level with an average elevation of 5 m.

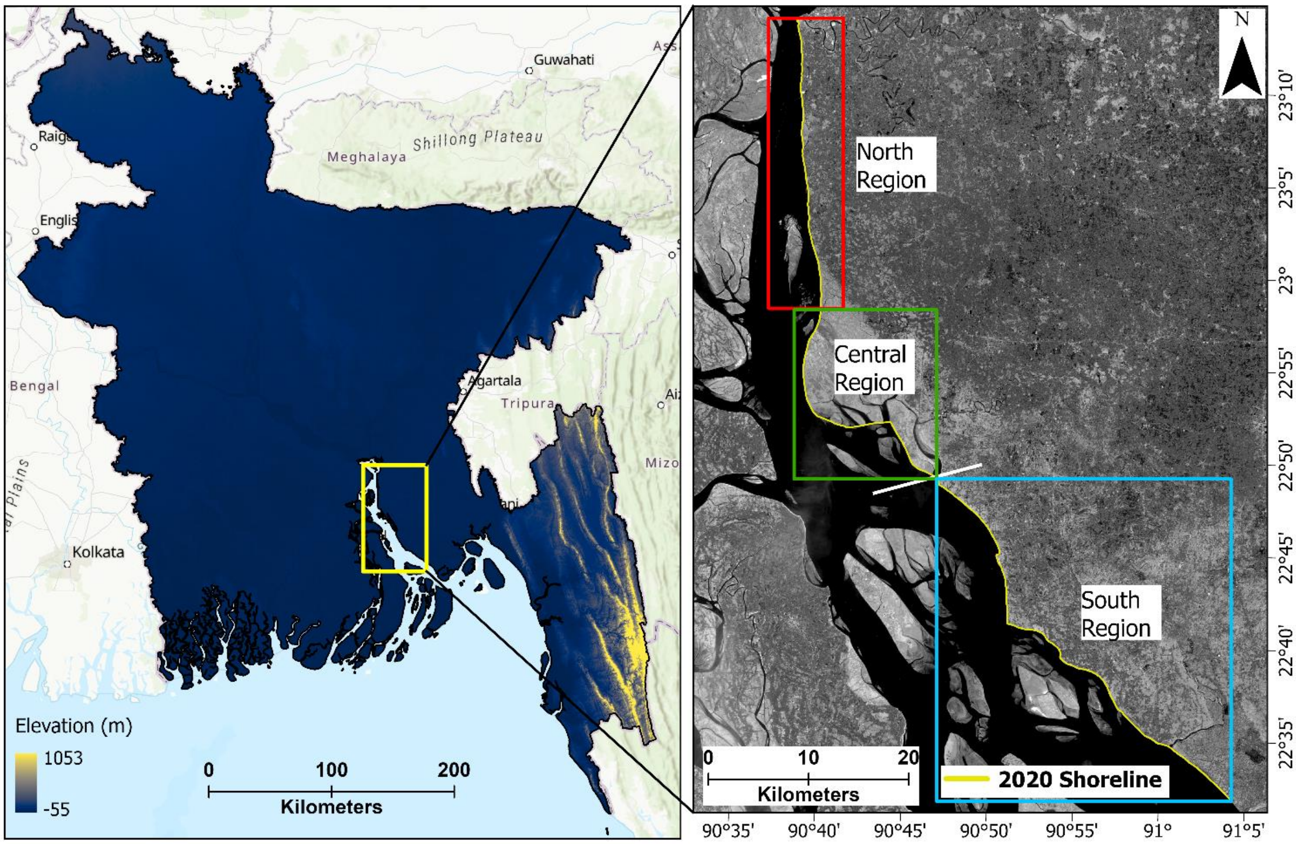

Figure 1. Study area - the eastern bank of lower Meghna River region in Bangladesh.

2.2. Data

This study analyzed satellite data for a period of 34 years from 1988 to 2021. Multi-temporal Landsat satellite data was obtained from the USGS Earth Explorer website (https://earthexplorer.usgs.gov/, accessed on 2 May 2021). Landsat imagery is widely used for quantification of coastal erosion [10,11,32,33]. In order to obtain cloud-free imagery, satellite images were selected and downloaded for the dry season winter months (December to February) every year. A detailed description of the imagery used can be found in Table A1 of Appendix A.

2.3. Methods

In this study, shoreline prediction performance was analyzed using the aforementioned Landsat time-series data of 1988–2021. Satellite images were processed using ArcGIS Pro software. The Digital Shoreline Analysis System (DSAS) was used to calculate shoreline change rates, and simple extrapolation and the Kalman filter method were used to predict future shoreline positions. First, images were preprocessed using ArcGIS Pro. Second, annual shoreline positions were extracted (e.g., vector shorelines) and shoreline uncertainties were assessed. Third, using DSAS, shoreline change rates were calculated. Finally, shoreline change rates were used to predict future shoreline positions and prediction accuracy was assessed. A detailed diagram of the work flow used in this study can be found in Figure 2.

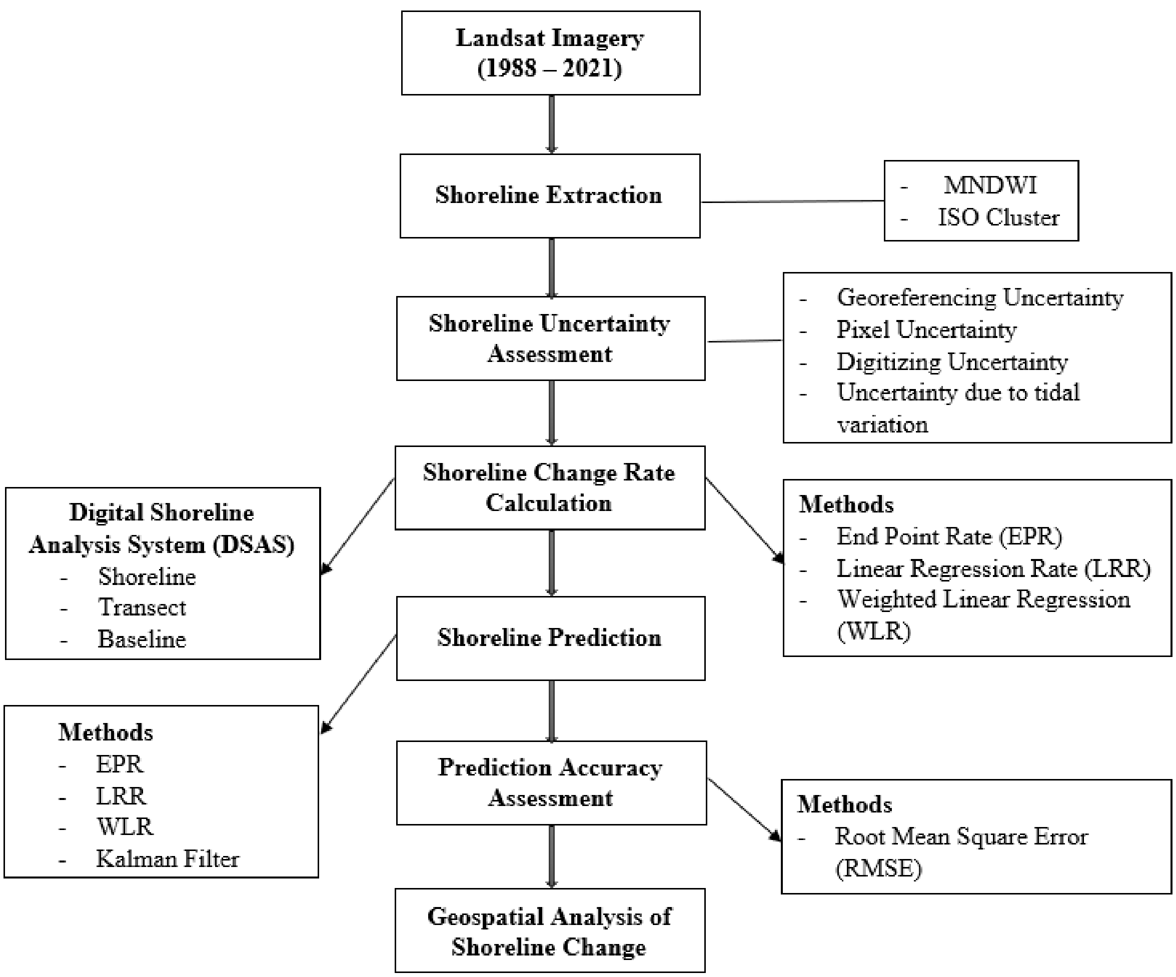

Figure 2. Flowchart of the methodology followed to conduct this study.

2.3.1. Shoreline Extraction

To extract annual shorelines, the Modified Normalized Difference Water Index (MNDWI) were used as an input with other Landsat bands. The middle infrared (MIR) and green bands are used to create this index which enhances the ability to distinguish between open water features while suppressing/removing other noises, such as built-up land, vegetation, and soil noise [34]. MNDWI is defined as:

$$\mathrm{MNDWI} = (\mathrm{Green} - \mathrm{MIR})/(\mathrm{Green} + \mathrm{MIR}) \tag{1}$$

The MNDWI value ranges from −1 to 1, where water pixels approach to 1, and are easily distinguishable from other pixels. It is a widely used method to separate water and non-water pixels. After getting the index, the ISO cluster unsupervised classification was used to classify the images into 10 classes. In this process, we used red, green, blue, and MIR bands, along with MNDWI as an input. After obtaining the initial ISO cluster classification output layer, the output layers were reclassified into two classes, water and non-water via visual interpretation. The land–water interface defining shoreline was then converted to a vector digital shoreline employing a smoothing algorithm. Results yielded annual dry season shorelines for every year from 1988 to 2021.

2.3.2. Shoreline Uncertainty

The accurate estimation of shoreline errors and uncertainties is necessary. The accuracy of the shoreline movement estimation is affected by several factors. This study assessed and accounted for four uncertainty terms provided by Hapke et al. [35]. Shoreline uncertainty was assessed using the following equation:

$$\text{Utotal} = \sqrt{Ug^2 + Up^2 + Ud^2 + Ut^2} \quad (1)$$

where, *Utotal* = total uncertainty, *Ug* = georeferencing uncertainty, *Up* = pixel uncertainty, *Ud* = digitizing uncertainty, and *Ut* = uncertainty associated with tide.

Georeferencing uncertainty (Ug) of the Landsat imagery used varied from 3.5 m to 7.3 m. This information was obtained from the metadata provided with the imagery. Pixel uncertainty (Up) is related to the pixel resolution of the imagery. All imagery we used had a 30 m resolution. That said, the pixel uncertainty (Up) was 30 m for all the images used in this study. The uncertainty related to the manual digitizing of shorelines from visual inspection is typically defined as digitizing uncertainty (Ud). Since we used the automated shoreline digitization method, our digitizing uncertainty (Ud) value is 0. To assess uncertainty due to tidal variation (Ut), we adopted the same approach used in our previous research [12]. Six different shorelines were derived for the dry season of 2000 using Landsat imagery. We found that the tidal levels ranged from 0.8 m to 2.8 m for the dates/times of the analyzed images. To measure the shoreline position difference, depending on the tide level, fifteen different combinations of shorelines for each transect were quantified. Our calculation suggested that the mean tidal difference was 0.97 m. We used regression analysis to reveal the shoreline uncertainty. Our analysis of shoreline uncertainty exhibits 1 m tidal difference, resulting in a 7 m shoreline shift. As a result, we applied a tidal uncertainty of 7 m. A detailed summary of the shoreline uncertainty assessment can be found in Table 1.

Table 1. Summary statistics of different parameters used in shoreline uncertainty assessment.

	Georeferencing Uncertainty (*Ug*)	Pixel Uncertainty (*Up*)	Digitizing Uncertainty (*Ud*)	Tidal Uncertainty (*Ut*)	Total Uncertainty (*Utotal*)
Minimum	3.54	30	0	5.60	30.72
Maximum	7.34	30	0	19.60	36.58
Mean	4.72	30	0	7.32	32.27

All units are in meters.

2.3.3. Shoreline Change Rates Calculation

Shoreline change rates were calculated using the DSAS (Digital Shoreline Analysis System), a software package for the quantification of shoreline movement developed by United States Geological Survey [25]. The EPR, LRR, and WLR were used to calculate the shoreline change rates at a 95% confidence interval. The EPR rate is calculated by dividing the distance of shoreline movement by the time period between the oldest and the most recent shoreline [29], yielding a change rate in units of meters per year. EPR is one of the most widely used methods for shoreline change rate calculation [23,36]. The LRR rate is calculated by fitting a least-squares regression line to all shoreline points for digital transects that are cast orthogonal from an offshore or onshore baseline to the mapped shorelines. With the x-axis being time (e.g., year), and the y-axis being the distance from the baseline to the observed shorelines, the regression coefficient and its uncertainty is an estimate of the true shoreline change rate and its associated uncertainty. This method is applied to the set of transects such that each transect obtains a transect-specific shoreline change rate and uncertainty value. In practice, it is common to use a transect spacing of 50 m; the spacing adopted here resulted in 1551 transects spanning the ~85 km shoreline of the study areas. Similarly, the WLR rate is calculated by fitting a least-squares regression line to all shoreline points for each transect by taking into consideration the uncertainty values associated with measurement [29]. In a weighted linear regression, the more reliable data are given greater emphasis or weight towards determining a best-fit line. The weight (w) is defined as a function of the variance in the uncertainty of the measurement (e).

$$w = 1/e^2 \qquad (2)$$

where, e is shoreline uncertainty value.

2.3.4. Shoreline Prediction and Temporal Strategy

For the shoreline prediction, we have used both simple extrapolation and Kalman filter methods, which take shoreline change rates as input, such as the EPR, LRR, and WLR. Prediction involves a general two-step process. For the simple extrapolation of historical rates, ArcGIS, via relevant geoprocessing tools, was used. We also calculated the advanced form of extrapolation using the Kalman filter method using the DSAS software. In short, for each of the 1551 digital transects, a “terminal shoreline position”, which is the x,y location of the intersection of the transect with an empirically mapped digital shoreline (e.g., shoreline derived from Landsat) for the year from which a future shoreline position prediction is made, is shifted either onshore or offshore depending on the empirically derived shoreline change rate which may be either positive or negative indicating shoreline accretion or erosion. In this way, a predicted future shoreline position along each transect for a defined time horizon into the future is produced.

Important to this research is the establishment of a temporal strategy to predict future shorelines. This is important to satisfy our goal and the innovation of evaluating how prediction performance varies over time and space. Our strategy required consideration of both the time depth of historical shorelines (e.g., how many years of annual input shorelines) used to quantify historically observed shoreline change rates and the time horizon for future prediction shorelines (e.g., how far out in the future to predict). We used time depths of 5, 10, and 15 years in the past and time horizons of 1, 5, 10, and 20 years into the future.

As described below, we used the LRR rates for all shoreline predictions using both the simple extrapolation and Kalman filter methods. In the case of Kalman filter prediction, we could not predict future shoreline positions using 5-year time depth data due to technical issues with the DSAS, though we were able to do so for 10 year and 15-year time depth data. A large dataset was generated in combination of four different methods of prediction, three different time depths and four different time horizons. Approximately 300k observations were produced at the transect level through the above combinations where digital transects at 50 m spacing were cast orthogonal from a digital baseline such that transects intersected the shorelines at approximately 90 degrees. Through our previous research [11] and personal observation, we found that the central zone of our study produced some noise in generating data due to the nature of the complexity of erosion and accretion in this region. As a result, we state that some transects at the central region will result in large rates of shoreline positional change that are statistical outliers. We, therefore, trimmed to remove these outliers from the data. Based on empirical data inspection, we excluded transects with change rates exceeding 500 m/year. Through this process, we removed approximately 2.4% of the observations from the final data analysis.

2.3.5. Prediction Performance Metric

Empirical prediction of future shorelines requires metrics to evaluate prediction accuracy. Prior research [21,26,27] has used the root mean square error metric (RMSE) to measure prediction accuracy, a common measure of accuracy assessment across science domains. The RMSE is the standard deviation of the residuals (prediction error). To assess the prediction accuracy, this study used RMSE as a performance metric. RMSE is applied to individual transects. The study area contains 1551 transects at a 50-m spacing that are roughly equally distributed into three regions following prior research of this study area [11], the north, central, and south zones. A basic building block of RMSE is to quantify the difference in meters between a predicted and an observed shoreline position. The difference in position along the transect between the predicted and observed shoreline position can be either negative or positive. RMSE is defined as:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \tag{3}$$

where, y_i is the actual shoreline position, $\hat{y}_i$ is the predicted shoreline position, and N is the total number of observations.

3. Results

3.1. Correlation of Different Shoreline Erosion Rates

In this study, the EPR, LRR, and WLR methods were used to calculate shoreline change rates. Figure 3 reveals patterns of spatio-temporal correlation of shoreline rates between different methods used. Our results suggest that among the methods used, the LRR and WLR rates are highly positively correlated (>99%) for all time depths and regions. Since the LRR and WLR rates are highly correlated, a similar pattern of correlation between EPR vs. LRR and EPR vs. WLR were observed. Our analysis also suggests that correlation values varied over time and space. The correlation between EPR and LRR rates ranged from 0.7-1 (5 y time depth), 0.62-1 (10 y time depth), and 0.54-1 (15 y time depth). Most of the years, the correlation between the EPR and LRR were found to be strong (>0.7) for all regions, although a small number of observations had moderate (0.5-0.7) correlation values. These fluctuations in correlation values might be related to the way EPR and LRR rates are calculated. The EPR rates use only the beginning and end year shorelines while calculating the shoreline movement. On the other hand, the LRR rates use all the available shorelines to calculate shoreline change rates.

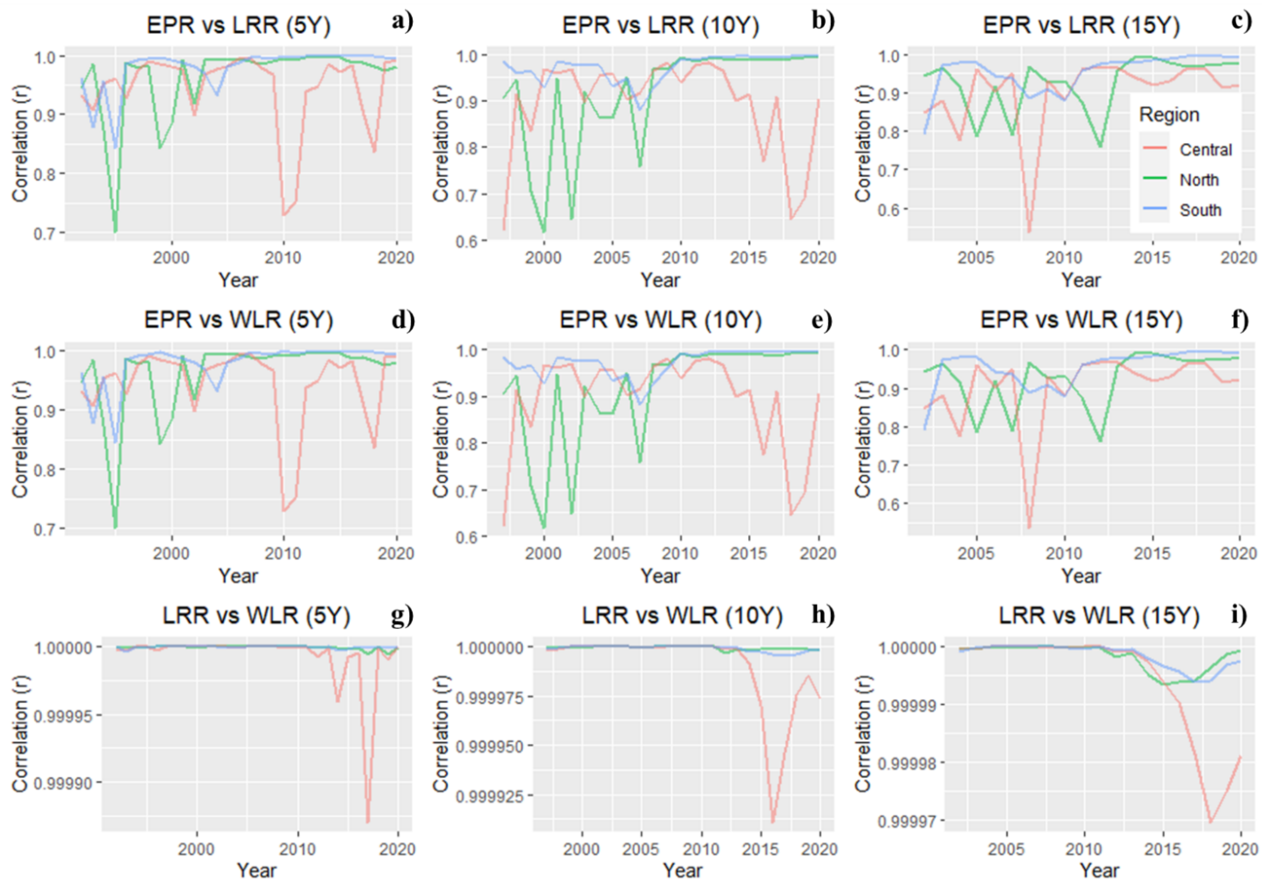

Figure 3. Spatio-temporal pattern of the Pearson correlation (r) values between different methods used to calculate historical shoreline change rates: (**a**) 5 years' time depth EPR vs. LRR, (**b**) 10 years' time depth EPR vs. LRR, (**c**) 15 years' time depth EPR vs. LRR, (**d**) 5 years' time depth EPR vs. WLR, (**e**) 10 years' time depth EPR vs. WLR, (**f**) 15 years' time depth EPR vs. WLR, (**g**) 5 years' time depth LRR vs. WLR, (**h**) 10 years' time depth LRR vs. WLR, and (**i**) 15 years' time depth LRR vs. WLR.

3.2. Spatio-Temporal Variation in Shoreline Change Rates

Figure 4 presents the LRR rates of shoreline movement over time and space. We present results calculated using the LRR rates but not the WLR rates due to the very high correlation of LRR and WLR. The LRR rates were also chosen over the WLR rates to align with the beta forecasting (Kalman filtering) used in the later part of the study to assess the prediction accuracy. Additionally, we excluded the EPR rates, since it only uses two specific shorelines to calculate the change rate. The LRR and WLR rates were found to be more sophisticated because they use all the shorelines available to calculate the rate.

Among the regions, north and south region experienced primarily erosion while central region experienced mostly accretion over the studied period (1988-2021) (Figure 4; Appendix B). The LRR rates suggest that the central region experienced the highest average accretion rates (Figure 4a) in the years 1996-1998. During these years, average accretion rates were 177 m. On the other hand, the north and south regions experienced erosion most of the years. The highest rates of erosion were 109 m for the north region and 173 m for the south region in the years 2011 and 2015, respectively.

Among the time depths of data used, the distribution of the erosion/accretion rates suggest that the 15 years' time depth data produced the lowest accretion/erosion rates for all regions compared to the 5 years and 10 years of data used in the model. This is likely due to the higher time depth smoothing out finer scale temporal variability compared to lower time depth. The highest rates of accretion (265 m) were found to be in the year 1998 when 10 years' time depth data used. On the other hand, the highest rates of average erosion (207 m) were found to be in the year 2014 for 5 years' time depth.

Figure 4. Linear regression rate of shoreline movement over time and space: (**a**) average of all time depths, (b) 5 years' time depth, (**c**) 10 years' time depth, and (d) 15 years' time depth. In the figure, positive values indicate erosion and negative values indicate accretion.

3.3. Spatio-Temporal Variation in Shoreline Prediction Performance

3.3.1. One-Year Shoreline Prediction Performance

Figure 5 reveals the spatio-temporal dynamics of 1-year prediction accuracy; e.g., predicting future shorelines one year into the future beyond a terminal date shoreline. The prediction accuracy of 1 year in the future was much better for the north and south regions compared to the central region. Our results suggest that among the regions studied, the RMSE values were lower in the recent years compared to the prior years for the north and south regions. On the other hand, the RMSE values fluctuated (62-2065 m) over time starting from the year 2007 for the central region. For this region, the prediction performed better in the prior years (before

2007) compared to the recent years. Due to the noisy nature of RMSE values for the different years in the central region, we have excluded the line showing the fluctuations in RMSE values. Removing the lines for central region in the plots (Figure 5) help in comparing with the RMSE values in the other figures.

In comparison of the RMSE values for different time depths used, this study found that the 5 years' time depth data performed best in the south region while 10- and 15-years' time depth data performed well in the north region. The average RMSE value was lowest (114 m) for the north region, while 15 years of time depth data used. The highest average RMSE values were found for the central regions, which can be explained due to the complex, noisy, and more highly variable nature of this region's shoreline. The average RMSE values ranged from 556-640 m for this region. While combining the RMSE values for the different regions using all time depths, this study found that the north region had the lowest error (147 m) in predicting 1 year in the future. On the other hand, the highest RMSE value (585 m) was found for the central region.

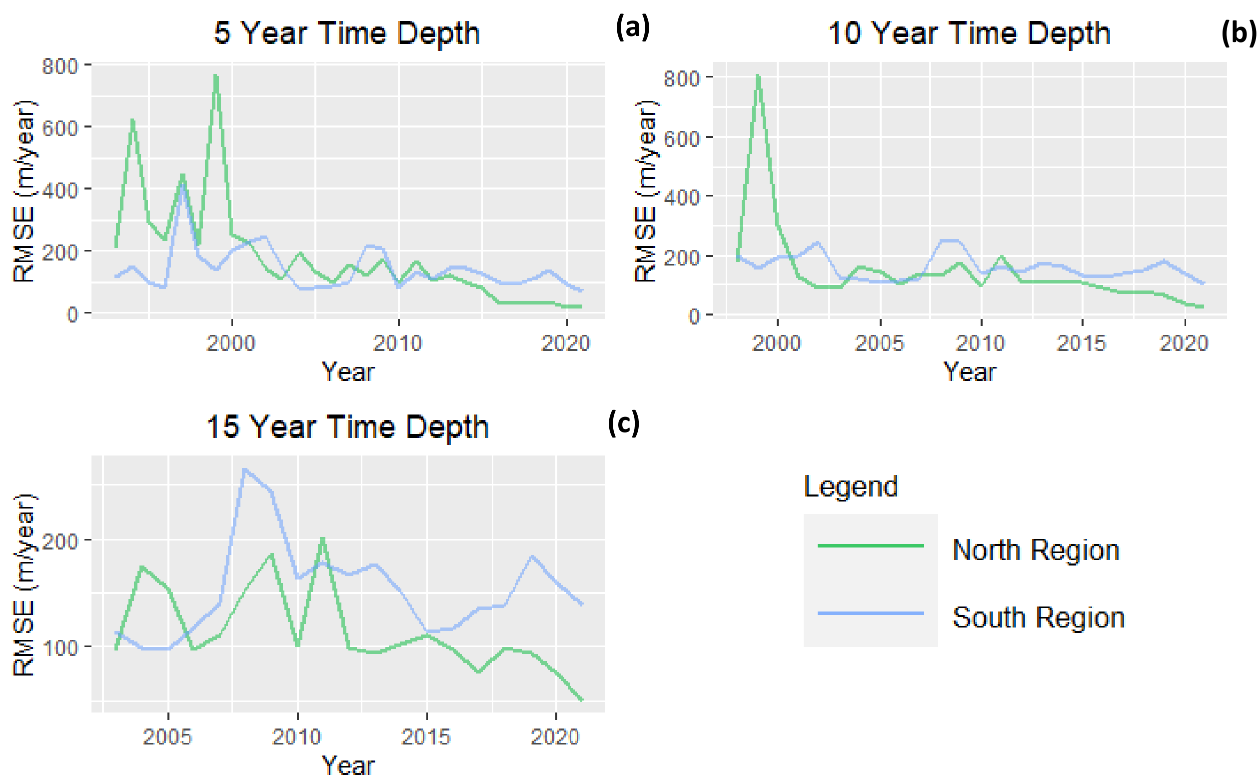

Figure 5. Variations in RMSE values calculated using the LRR rates data for predicting 1 year in the future: (**a**) 5 years' time depth, (**b**) 10 years' time depth, and (**c**) 15 years' time depth.

3.3.2. Five-Year Shoreline Prediction Performance

Similar to 1-year prediction, the LRR rates were utilized to predict shoreline positions 5 years into the future. Figure 6 illustrates the spatio-temporal pattern of the shoreline prediction performance using RMSE values. The prediction performance was again much better for the north and south regions compared to the central region for all different time depths used in this study. We believe that the RMSE values were much higher for the central region due to its complex nature and higher accretion rates over time.

We found that the north region had the lowest average RMSE value (111 m/year) over other the two regions followed by the south region (137 m/year) while combining all time depth data. The average RMSE value of different time depth data suggest that the north region had the lowest average RMSE value (101 m/year) using 15 years' time depth. The aggregation of all regions' RMSE values suggest that the lowest (194 m/year) and highest (208 m/year) values were found when 5 years' and 15 years' time depth data was used, respectively.

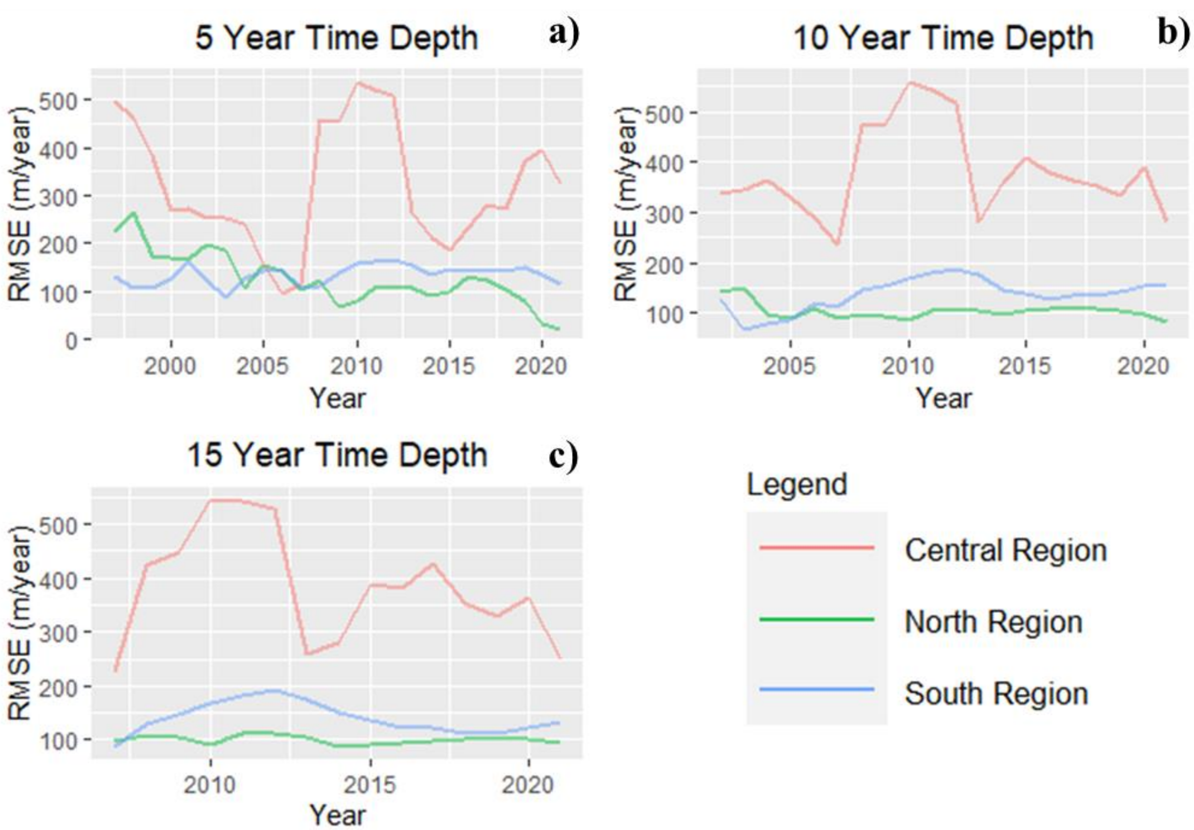

Figure 6. Variations in RMSE values calculated using the LRR rates data for predicting 5 years in the future: (**a**) 5 years' time depth, (**b**) 10 years' time depth, and (**c**) 15 years' time depth.

3.3.3. Ten-Year Shoreline Prediction Performance

Using normal extrapolation and the Kalman filter method, shoreline positions were predicted for 10 years into the future and accuracy was assessed for all regions and different time depths (Figure 7). Our results suggest that, irrespective of the methods and time depths of the data used, the north and south regions performed better in predicting future shoreline positions compared to the central region. As already mentioned in the previous sections, this might be due to the complex nature of the shoreline movement in the central region.

In comparison between the LRR rates and Kalman filter predictions, our analysis suggests that the LRR rates outperformed the Kalman filter method of prediction. The RMSE values were found to be much lower when the LRR rates were used in prediction. While using the LRR rates for all depths, the lowest average RMSE value was found to be 106 m/year for the north region, while the highest value was found to be 252 m/year for the south region. On the other hand, for the Kalman filter method, the lowest and highest average RMSE value was 197 m/year and 503 m/year for north and south region respectively.

In terms of time depths used, the prediction accuracy was found to be much higher for the increased number of shorelines used for both the LRR rates and the Kalman filter method. The lowest average RMSE value was found to be 149 and 260 m/year for the LRR rates and Kalman filter method while 15 years of shoreline data was used in the model. This clearly indicates that higher numbers of shorelines yield higher accuracies. Results also indicate that the simple extrapolation outperformed than the Kalman filter method.

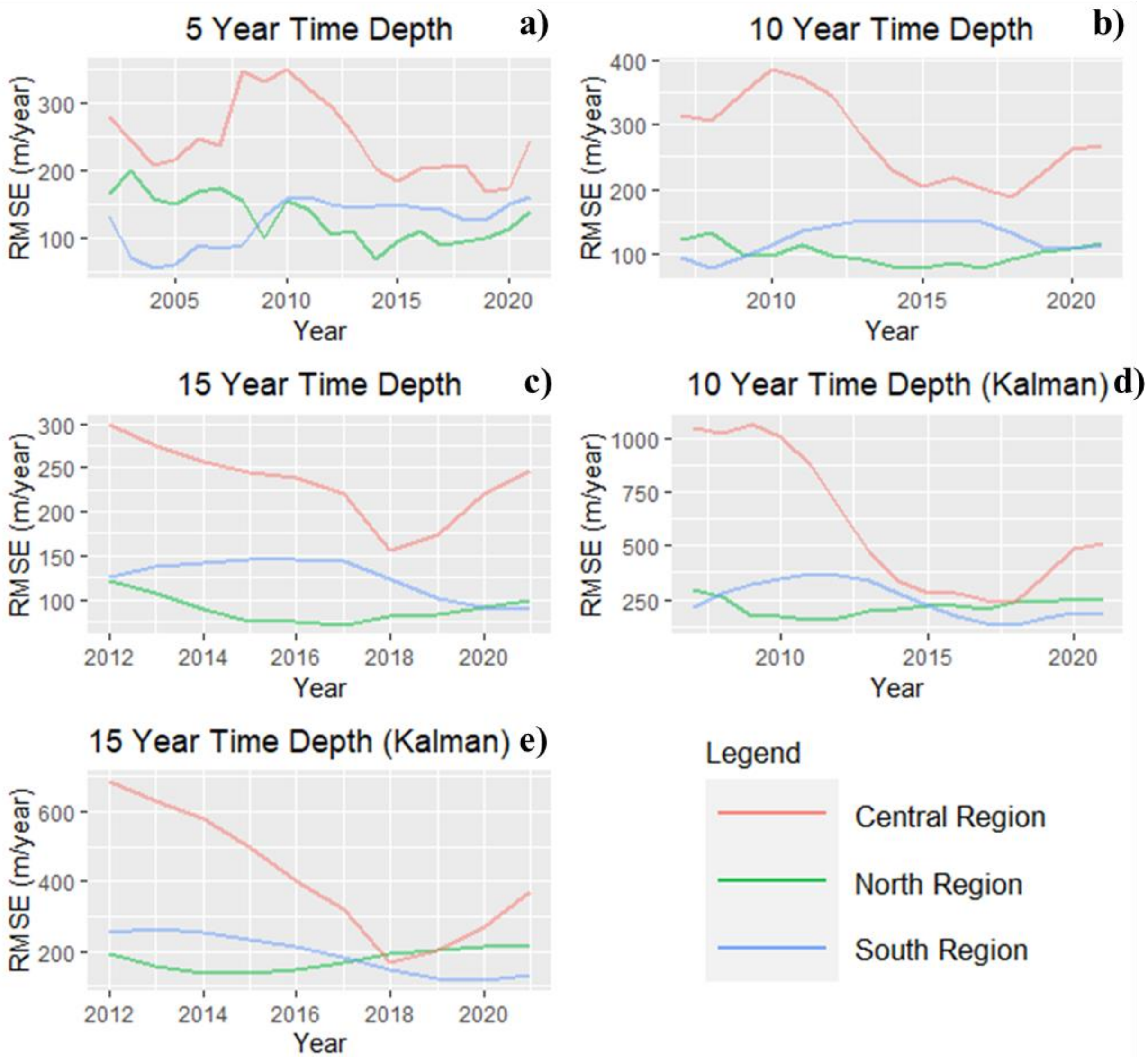

Figure 7. Variations in RMSE values of the LRR rates in predicting 10 years in the future calculated: (**a**) 5 years' time depth, (**b**) 10 years' time depth, (**c**) 15 years' time depth, (d) 10 years' time depth (Kalman), and (**e**) 15 years' time depth (Kalman).

3.3.4. Twenty-Year Shoreline Prediction Performance

Figure 8 shows the spatio-temporal dynamics of shoreline prediction accuracy for 20 years in the future. Similar to the 10 years' prediction, both the LRR rates and Kalman filtering methods were used to predict shoreline positions. Like 10 years prediction, simple extrapolation performed better than Kalman filter prediction. Results show that the average RMSE value of the LRR rates was 161 m/year while the value for Kalman filter method was 321 m/year.

Among the regions, and for the other different time horizons, the north and central regions performed better in predicting future shoreline positions. The average RMSE for the north, central, and south regions were found to be 123, 257, and 161 m/year, respectively, when the LRR rates were used. With the Kalman filter prediction, it was 150 m/year for the north, 647 m/year for the central, and 165 m/year for the south region.

The average RMSE value for 5 years' time depth (156 m/year) were found to be the lowest among all depths used in predicting shorelines 20 years in the future. In comparison, between simple extrapolation and the Kalman filter prediction, it was found that the RMSE value was almost doubled for the Kalman method compared to the LRR rates. This result clearly suggests that the normal extrapolation outperformed the Kalman filter method in predicting 20 years into the future.

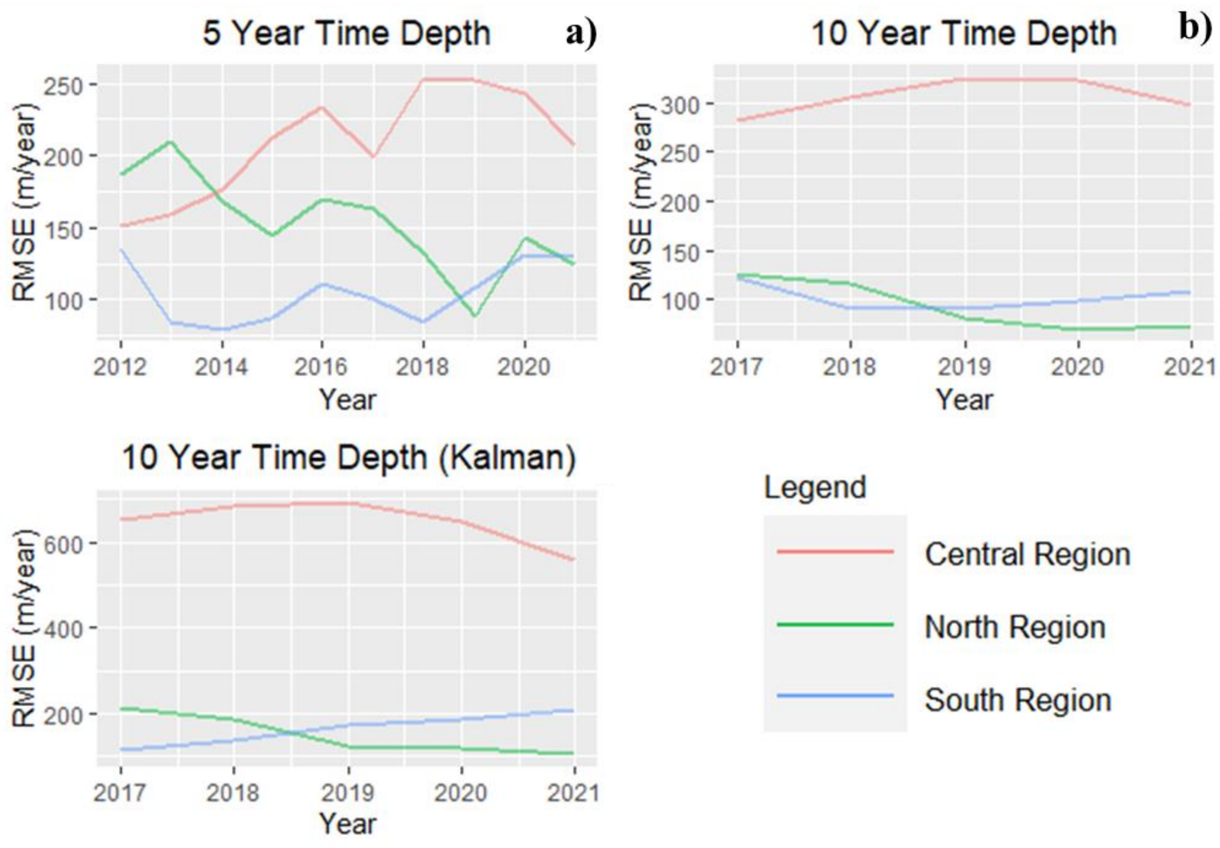

Figure 8. Variations in RMSE values calculated using the LRR rates data for predicting 20 years in the future: (**a**) 5 years' time depth, (**b**) 10 years' time depth, and (**c**) 10 years' time depth (Kalman).

4. Discussion

In this study, the performance of future shoreline predictions was assessed using satellite imagery from 1988 to 2021. Approximately 85 km of the shoreline along the eastern bank of the lower Meghna River region in Bangladesh was used as the case in this study. The entire shoreline was divided into three zones based on their distinct rates of shoreline erosion and accretion. Previous research found that the north and south regions experienced extreme erosion over the last three decades, while the central region experienced dominant accretion [11,37], a finding we replicate here.

Our results suggest that the performance of shoreline predictions varied over time and space. It also varied by different methods used to predict future shoreline positions. The predictions performed best with relatively higher number of shorelines used as input to derive shoreline change rates. A clear finding is that higher time depth in estimating the shoreline change rates is preferable. Our analysis of RMSE for the prediction of future shoreline positions suggests that prediction performed better for the near future compared to distant. To come up with this conclusion, we compared the average RMSE values for the north and south regions (all time depths) of the 1 and 5 years (near future) to the mean of the 10- and 20-year values (distant future). Our results suggest that the average RMSE value for near future (138 m/year) was lower than the distant future (152 m/year). Thus, prediction for shorter time horizons is preferable, at least for this region and the statistical empirical methods employed. Among our study's three defined regions, the north and south regions performed better in predicting future shoreline positions compared to the central region. This is understandable due to the complex and generally "noisy" form in terms of shoreline data mapping in the last decade in the central region of the study area. Our results also suggest that the simple extrapolation outperformed the Kalman filter method for the future shoreline prediction. The average RMSE values for simple extrapolation (128 m/year) were found to be much lower than the Kalman filter method (185 m/year).

Many studies have used DSAS to predict future shoreline positions. However, very few studies have assessed the performance of different methods used in predicting shoreline movement with the relatively rich temporal time series employed here. A recent study by Ciritci and Turk [21] used the DSAS model to compare the prediction accuracy of the Kalman filter and other statistical methods, such as the EPR, LRR, and WLR, using a future prediction time horizon of 10 and 20 years. They found that while using the Kalman filter method, the 10-year shoreline prediction provided a higher accuracy over the 20-year prediction. This result is aligned with our findings that the accuracy of the prediction is generally higher for the near future compared to the distant. They also found that among the EPR, LRR, and WLR methods, WLR provided more accurate results compared to the others. However, our finding suggests that the Kalman filter method underperformed compared to the simple extrapolation runs counter to prior findings. The possible reason behind this might be related to the nature of the shoreline and the severity and magnitude of the erosion rates in this region. Future research might analyze how the prediction performance varies over different regions with different shoreline change rates.

Compared to much of the coastal erosion literature, this work drew from a 34-year time series of satellite-derived shorelines at annual temporal resolution. This time depth enabled us to employ a temporal design strategy expected to yield a robust characterization of the space-time erosion patterns. It also enabled us to empirically assess the performance of future shoreline position predictions by quantifying how prediction performance varies depending on the time depths of the input historical shoreline data and the time horizons of predicted shorelines.

Though the location of this study is coastal Bangladesh, this is a methodological innovation with the potential for forecasting applications to other deltas and vulnerable shorelines globally. While empirical results are specific to the project's study area, results can inform the region's shoreline forecasting ability and the associated mitigation and adaptation strategies. This study also has implications in better planning and management of the lower Meghna River region of Bangladesh.

This study used the empirical data derived from satellite imagery to predict future shoreline positions. However, shorelines are heavily influenced by sediment transportation, local rainfall and floods, cyclones, and sea level rise. It is predicted that due to global warming and the resultant sea level rise, coastal erosion will increase in the future [38]. Since this study did not consider these influences in predicting future shoreline positions, this study has limitations. Further study is needed to consider the influence of the above-mentioned factors and to engage more deeply with comparison of the simple extrapolation method vs. the Kalman filter method for predicting future shorelines. While the empirical predictive methods used in this study may not be able to account for the physically process-based factors contributing to coastal erosion and accretion, they may be useful to inform both local residents and policy makers to plan of future impacts of shoreline change [39].

Due to the extreme rates of shoreline change in this region, we believe the errors are seen to be generally larger than the other similar studies around the world. Due to high errors, this kind of method might not be well suited to predicting shorelines in this region. However, we found that the error was much lower for the northern and southern regions compared to the central region. Further studies might need to assess how the prediction performance varies based on the shoreline change rates. A study on the Puri Coast of India by Mukhopadhyay et al. [40] found that the RMSE value to be 51.34 m when EPR rates were used. However, they didn’t mention the actual rate of erosion. Another study in the same coast found that the mean erosion rate was 16.33 m/year during 1990-2015 [41]. This indicates that the RMSE value was much higher for this coast compared to the actual erosion rates. It might be related to the long-term prediction of the coastal shoreline movement. As such, we suggest predicting for the near future (e.g., 1 year, 5 years) due to the high error in future shoreline prediction.

The tools used in this study do come with limitations. The DSAS tool used for future shoreline positions may not be ideal for all data types, locations and patterns of shoreline change [22,29]. Future researchers and those in applied practice should be cognizant of the methodological limitations when using competing shoreline change rates and future shoreline prediction models. A clear message and contribution of this work is that shoreline change rates and the resulting future shoreline positional prediction can vary depending on: (1) choice of change rate used, (2) choice of time depth used to derive change rates, (3) choice of time horizon for predicting future shorelines, (4) the time period in which rates are derived and future shoreline positions are predicted, and (5) the regional locations for which rates, predictions, and time periods are employed. In conclusion, both methodological considerations of rates and prediction methods along with considerations of time and space variability must be considered.

5. Conclusions

This study was carried out to assess the performance of empirical forecasting of future shoreline positions. To do so, this study used the eastern bank of the lower Meghna River in Bangladesh as the case site. This area is well evident to be one of the most erosion-prone areas in the world. This study adopted a data-driven approach to predict future shoreline positions using various statistical methods by examining past shoreline change rates at a 95% confidence interval (CI). Using different time depths of shoreline rates, methods of erosion rate calculation, and predicting future shoreline positions for different time horizons, this study assessed the prediction performance of different combinations. This study concludes that the higher the number of shorelines used in calculating and predicting shoreline the better the performance is. It was revealed that the prediction accuracy is generally higher for the near future (138 m/year) compared to the distant future (152 m/year). Our results also suggest that the normal extrapolation method (128 m/year) of prediction outperformed the Kalman filter method (185 m/year) in predicting future shoreline positions. However, it is important to determine the historical change rates and predict future shoreline positions for the long-term planning and management of coastal areas. It is critically important because future climate change is likely to exacerbate coastal erosion globally.

Appendix A

Table A1. Datasets used in the study.

Image Date	Representative Year	Satellite	Path/Row	Resolution (m)
19 February 1988	1988	Landsat 5	137/44	30
21 February 1989	1989	Landsat 5	137/44	30
7 January 1990	1990	Landsat 5	137/44	30
15 March 1991	1991	Landsat 5	137/44	30
12 December 1991	1992	Landsat 5	137/44	30
30 December 1992	1993	Landsat 5	137/44	30
2 January 1994	1994	Landsat 5	137/44	30
21 January 1995	1995	Landsat 5	137/44	30
9 February 1996	1996	Landsat 5	137/44	30
26 January 1997	1997	Landsat 5	137/44	30
14 February 1998	1998	Landsat 5	137/44	30
17 February 1999	1999	Landsat 5	137/44	30
20 February 2000	2000	Landsat 5	137/44	30
5 January 2001	2001	Landsat 5	137/44	30
1 February 2002	2002	Landsat 7	137/44	30
2 December 2002	2003	Landsat 7	137/44	30
13 December 2003	2004	Landsat 5	137/44	30

15 December 2004	2005	Landsat 5	137/44	30
4 February 2006	2006	Landsat 5	137/44	30
22 January 2007	2007	Landsat 5	137/44	30
16 December 2007	2008	Landsat 7	137/44	30
3 January 2009	2009	Landsat 7	137/44	30
30 January 2010	2010	Landsat 5	137/44	30
1 January 2011	2011	Landsat 5	137/44	30
28 January 2012	2012	Landsat 7	137/44	30
14 January 2013	2013	Landsat 7	137/44	30
24 December 2013	2014	Landsat 8	137/44	30
25 November 2014	2015	Landsat 8	137/44	30
30 December 2015	2016	Landsat 8	137/44	30
1 January 2017	2017	Landsat 8	137/44	30
4 January 2018	2018	Landsat 8	137/44	30
7 January 2019	2019	Landsat 8	137/44	30
10 January 2020	2020	Landsat 8	137/44	30
27 December 2020	2021	Landsat 8	137/44	30

Appendix B

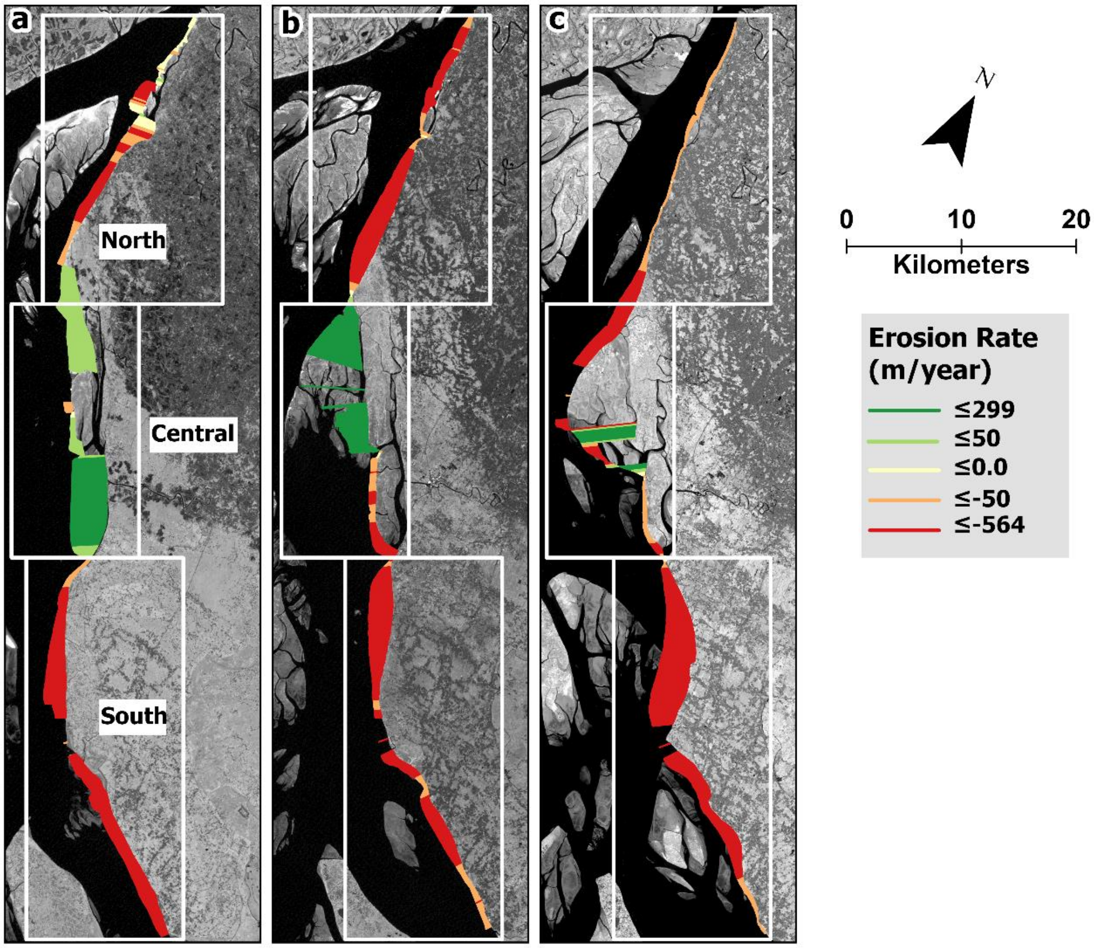

Figure A1. Decadal shoreline movement in the eastern bank of the lower Meghna River region in Bangladesh: (**a**) 1990 - 1999, (**b**) 2000 - 2009, (**c**) 2010 - 2019. In the map, positive values indicate accretion and negative values indicate erosion.

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Chapter 3. Evaluation of predicted loss of different land use and land cover (LULC) due to coastal erosion in Bangladesh

This chapter evaluates predicted loss of different land use and land cover (LULC) due to coastal erosion in Bangladesh. It was published as manuscript in the journal "*Frontiers in Environmental Science*".

Islam MS, Crawford TW and Shao Y (2023). Evaluation of predicted loss of different land use and land cover (LULC) due to coastal erosion in Bangladesh. *Front. Environ. Sci.* 11:1144686. doi: 10.3389/fenvs.2023.1144686

Abstract

Coastal erosion is one of the most significant environmental threats to coastal communities globally. In Bangladesh, coastal erosion is a regularly occurring and major destructive process, impacting both human and ecological systems at sea level. The Lower Meghna estuary, located in southern Bangladesh, is among the most vulnerable landscapes in the world to the impacts of coastal erosion. Erosion causes population displacement, loss of productive land area, loss of infrastructure and communication systems, and, most importantly, household livelihoods. With an aim to assess the impacts of historical and predicted shoreline change on different land use and land cover, this study estimated historical shoreline movement, predicted shoreline positions based on historical data, and quantified and assessed past land use and land cover change. Multi-temporal Landsat images from 1988 – 2021 were used to quantify historical shoreline movement and past land use land and land cover change. A time-series classification of historical land use and land cover (LULC) were produced to both quantify LULC change and to evaluate the utility of the

future shoreline predictions for calculating amounts of lost or newly added land resources by LULC type. Our results suggest that the agricultural land is the most dominant land cover/use (76.04% of the total land loss) lost over the studied period. Our results concluded that the best performed model for predicting land loss was the 10-year time depth and 20-year time horizon model. The 10-year time depth and 20-year time horizon model was also most accurate for agricultural, forested, and inland waterbody land use/covers loss prediction. We strongly believe that our results will build a foundation for future research studying the dynamics of coastal and deltaic environments.

Keywords: Coastal Erosion; Land Loss; Land Use and Land Cover; Lower Meghna River; Bangladesh

1 Introduction

Coastal erosion is one of the most prominent problems in coastal areas globally and the world's mega-deltas are heavily impacted (Woodroffe et al., 2006). Human life and natural environments are threatened by coastal erosion. Due to erosion, coastal lands are being swallowed by the seawater and people are forced to move inland. A recent global study found that the earth surface lost about 28000 km^2 land during 1984 – 2015, which is twice the surface land area gained ((Mentaschi et al., 2018). Coastal erosion is caused by many different physical (e.g., strong wave action, upstream discharge, river bathymetry) and anthropogenic factors (e.g., global warming, sea level rise). Soil properties such as texture and structure govern pore size distribution have an influence on soil’s erodibility. Studies suggest that soil moisture has substantial influence on water runoff and soil erosion (Wei et al., 2007; Lal et al., 2023). Due to coastal erosion, people living in the coast lose their valuable lands and properties. Climate change driven coastal erosion is posing extra risk for the coastal ecosystems. Increased frequency and severity of the events including storm, rainfall and flood, increased freshwater input to the marine systems and lengthening open water periods are influencing coastal erosion process in addition to the rising sea level (Sanò et al., 2011; Radosavljevic et al., 2016). Mean sea level rose 11–16 cm during 20th century (Hay et al., 2015; Dangendorf et al., 2017), possibly exacerbating coastal erosion globally. Land subsidence is another factor that is influencing coastal erosion risk. For instance, a study on land subsidence and its effects on coastal erosion in the Nam Dinh Coast of Vietnam found that the combined effects of land subsidence and relative sea level rise (SLR) were responsible for 66% of the observed rate of erosion (Nguyen and Takewaka, 2020). Resulting inundation from rising seas will heavily impact low-lying areas, impacting at least 100 million persons globally who live within one meter of mean sea level (Zhang et al., 2004). This is a tremendous threat for the people living in the coastal communities.

Coastal communities in Bangladesh are vulnerable to different natural hazards and disasters due to close proximity to the sea and sea level. Most parts of coastal areas in Bangladesh are within few meters of the sea level. Extreme erosion is one of the biggest problems threatening the livelihoods of coastal populations, especially the poor. Coastal Bangladesh is highly vulnerable to coastal erosion; land erosion and accretion rates are among the highest in the world (Brammer, 2014; Ahmed et al., 2018; Crawford et al., 2021). It is a recurring driver of homelessness impacting thousands of people annually. The coastal poor are the most susceptible to these impacts. A study by Kumar et al., (2022) suggested that a proper afforestation measures in the upper catchments of Brahmaputra basin might reduce soil erosion and subsequent impacts on the riverbank communities.

Bangladesh is considered as one of the most vulnerable countries to the impacts of climate change in the world, particularly sea-level rise (Nicholls et al., 2007; Sarwar and Khan, 2007; Davis et al., 2018). Moreover, future global warming and subsequent rising sea levels seriously threaten coastal populations in this part of the world. Additionally, as economically marginalized populations live in coastal Bangladesh, the magnitudes and severity of the affected people may affect the total economic growth of the country (Hossain et al., 2016). As a lower middle-class country like Bangladesh, with limited internal resources, it is hard to cope with catastrophic natural hazards like erosion and its related consequences (Poncelet et al., 2010). Additionally, disasters like coastal erosion create the acute problems of unemployment in coastal rural areas and thereby worsen the socio-economic condition of displaced people. Most of these displaced people become landless, which pushes them further into poverty and forces migration. Future climate change is more likely to impact the migration decisions of the people living in coastal Bangladesh in search of prosperous livelihoods (Call et al., 2017).

Due to coastal erosion, people living on the coast not only lose their valuable lands and houses but also infrastructure, communication systems, and most importantly their livelihood (Monirul Alam et al., 2017). A proper estimation of different types of land loss is crucial for better understanding of the impacts of shoreline movement. It is particularly important for proper planning and implementation of appropriate environmental practices. Land use and land cover analysis using satellite imagery is a cost-effective way to monitor long-term changes. Many different methods are used to classify land use and land cover. The maximum likelihood, random forest, support vector machine, and artificial neural network methods were found to be most applied method for LULC analysis (Otukei and Blaschke, 2010; Rodriguez-Galiano et al., 2012; Balha and Singh, 2022). It is well evident that in the recent decades, human activity has significantly influenced land use and land cover globally (Lambin and Geist, 2008). A significant portion of the global population lives near the coast. As a result, it is expected that the coastal ecosystem will change globally due to a large number of human footprints in the coastal region. For regions experiencing the loss or gain of land resources due to erosion/accretion processes, it is important to map existing land use and land cover (LULC) and to integrate resulting data products with information on future shoreline change.

Coastal landscapes are changing rapidly due to both natural and anthropogenic activities in Bangladesh and elsewhere (Muttitanon and Tripathi, 2005; Olaniyi et al., 2012; Abdullah et al., 2019; Ekumah et al., 2020). In Bangladesh, coastal landscape is changing over time due to many reasons. A recent study by Abdullah et al., (2019) found that the vegetation coverage in the coastal Bangladesh is declining day by day due to natural (e.g. flooding, cyclones etc.) as well as human influenced alterations (e.g. shrimp farming). They also found that river areas have increased by 4.52 percent within 27 years (1990 – 2017). This indicates that, as this part of the country is the heart of the Ganges-Brahmaputra Meghna (GBM) delta, coastal areas of Bangladesh are experiencing erosion at alarming rates over the last few decades. Global warming and sea level rise is putting extra fuel on ever changing coastal landscape in Bangladesh. A similar study by (Rahman et al., 2017) found 26 percent increase in shrimp farm area from 1989 – 2015. During this 26 year of period, about 21% of bare lands declined. This is an indication that this highly dynamic coastal region of Bangladesh is changing rapidly.

With an aim to assess the impacts of historical and predicted shoreline change on different land use and land cover, this study estimated historical shoreline movement, predicted shoreline positions based on historical data, and quantified and assessed land loss due to historical and predicted shoreline movement. In our previous study, we assessed the predicted performance of shoreline movement (Islam and Crawford, 2022). We analyzed how prediction performance varies depending on the time depths of input historical shoreline data and the time horizons of predicted shorelines. Our results suggested that the higher the number of shorelines used in calculating and predicting shoreline change rates the better predictive performance was yielded. Though the prediction performance varied spatially, we found that prediction accuracies were substantially higher for the immediate future years compared to the more distant future. In this study, we assess how different LULC classes changed due to historical and predicted shoreline movement. The specific goals of this study were to 1) assess historical land loss in the Lower Meghna river region of Bangladesh, 2) measure the amount of different LULC has been lost to erosion over defined time intervals, and 3) estimate how well predicted shorelines predict amounts of succeeding LULC resources lost to erosion.

2. Materials and Methods

2.1 Study Area

The area covered in this study consists of Ramgati and Kamalnagar upazila of Lakshmipur district in Bangladesh. The upazilas are the second lowest elevation administrative units in Bangladesh. The study area covers approximately 885 km^2 with an estimated population of 490,000 (BBS, 2011). This area is located in coastal Bangladesh along the Lower Meghna River (Figure 1), which is highly prone to different natural hazards and disasters including coastal erosion, flooding, cyclones etc. This area is a major hotspot of coastal erosion in the country (Crawford et al., 2021). Peoples living in this part of the coastal Bangladesh are mostly dependent on agriculture and fishing for their livelihood (Paul et al., 2021; Rahman et al., 2021, 2022).

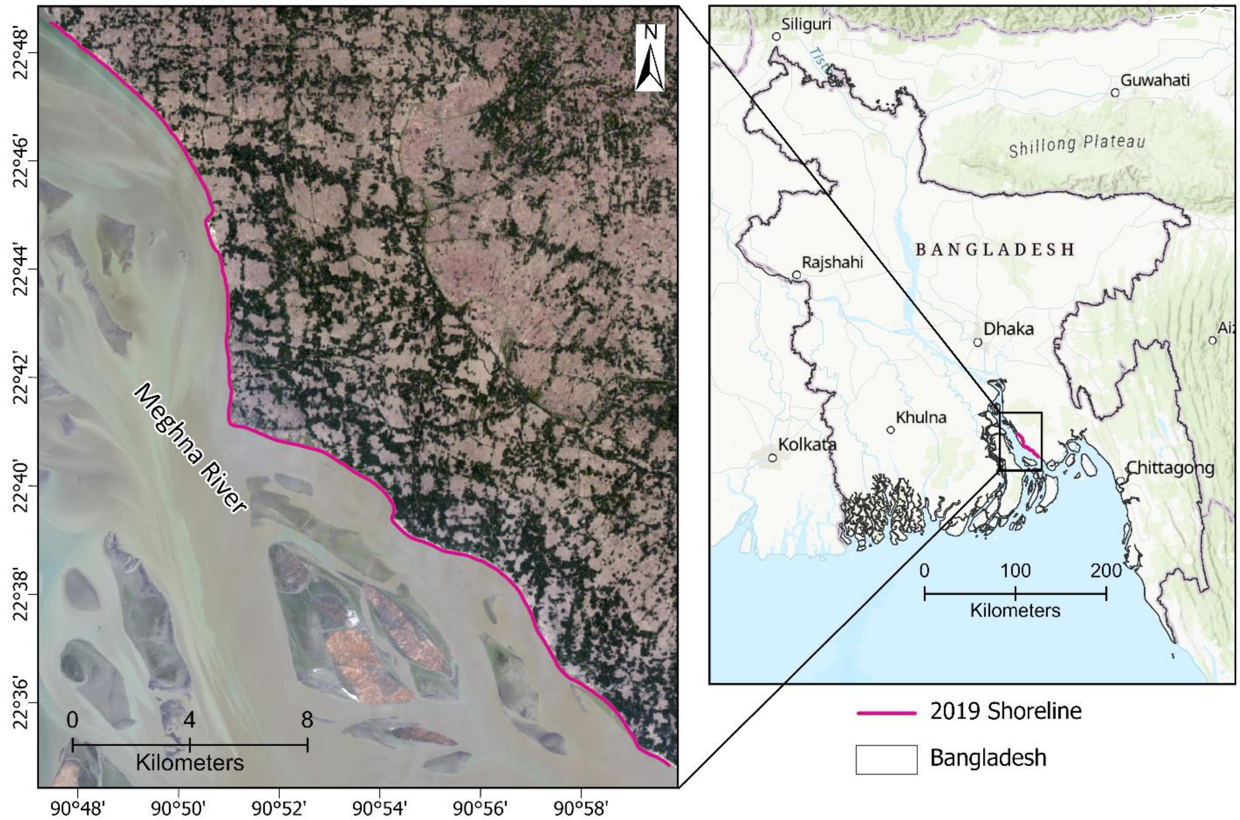

Figure 1. Map of the study area – lower Meghna River region of Bangladesh

2.2 Data

Multitemporal Landsat satellite imagery from 1988 – 2021 was used in this study. Landsat satellite data were obtained from the USGS Earth Explorer website (https://earthexplorer.usgs.gov/). Landsat imagery (30m resolution) were used for both shoreline mapping and LULC classification. To avoid cloud issues, we obtained imagery only from the dry season. This allowed us to acquire cloud free scenes. Landsat imagery is widely used for quantification of coastal erosion and land use and land cover change (Ghoneim et al., 2015; Ahmed et al., 2018; Crawford et al., 2021). A detailed description of the satellite images used in this study can be found in Table A1. Additionally, this research used high resolution satellite imagery from Google Earth Pro for post classification accuracy assessment.

2.3 Shoreline Extraction

After getting the imagery, multiple steps were followed to extract shorelines. First, the Modified Normalized Difference Water Index (MNDWI) was used to create an index that enhances the ability to distinguish between open water and non-open water sources (Xu, 2006). MNDWI makes use of the middle infrared and green bands. Values ranges from -1 to 1, where water pixels approach 1 and are easily distinguishable from other pixels. It is a widely used method to separate water and non-water pixels (Singh et al., 2015; Crawford et al., 2021). Second, MNDWI images were used as an input with other Landsat bands to classify images into different classes. The ISO cluster unsupervised classification was used to classify the image into 10 classes. After getting the classification output, it was reclassified into two classes, water and non-water via visual interpretation. A similar approach was used to generate annual vector shorelines from 1988-2021. Shoreline uncertainty was assessed by considering different uncertainty terms, including georeferencing uncertainty, pixel uncertainty, digitizing uncertainty, and uncertainty due to tidal variation. Shoreline total uncertainty was assessed using the following equation provided by (Hapke et al., 2011).

$$\text{Utotal} = \sqrt{Ug^2 + Up^2 + Ud^2 + Ut^2} \qquad (1)$$

Where, Utotal = total uncertainty, *Ug* = georeferencing uncertainty, *Up* = pixel uncertainty, *Ud* = digitizing uncertainty, and *Ut* = uncertainty associated with tide.

To assess shoreline uncertainty, we adopted a similar approach used in our previous research (Crawford et al., 2020; Islam and Crawford, 2022). The shoreline uncertainty analysis suggest that the mean uncertainty related to georeferencing (*Ug*), pixel (*Up*), digitizing (*Ud*), and tidal variation

(*Ut*) were 4.72, 30, 0, and 7.32m respectively. The mean total uncertainty (*Utotal*) was found to be 32.27m.

2.4 Shoreline Change Rates Calculation

Shoreline change rate was calculated using the DSAS (Digital Shoreline Analysis System), a software package for quantification of shoreline movement developed by United States Geological Survey (Thieler et al., 2009). There are several widely accepted methods for estimating coastal erosion rates including End Point Rate (EPR), Linear Regression Rate (LRR), Net Shoreline Movement (NSM), and Weighted Linear Regression (WLR)(Ciritci and Türk, 2020; Crawford et al., 2020, 2020). In this study, the LRR rates were used to estimate shoreline change rates. This method is the most used method for coastline change analysis (Kanwal et al., 2022). The LRR rate is calculated by fitting a least-squares regression line to all shoreline points for along digital transects that are cast orthogonal from an offshore or onshore baseline to the mapped shorelines. This method is applied to the set of transects such that each transect obtains a position-specific shoreline change rate and uncertainty value.

2.5 Shoreline Prediction

We used LRR rates to predict future shoreline position. Prediction involves a general two-step process. First, we generated shoreline change rates using the LRR method. Second, we used these rates as inputs for future shoreline prediction. Future shoreline prediction is implemented by simple extrapolation of historical rates deployed in ArcGIS via relevant geoprocessing tools and/or original scripts to automate processing. One of the essential aspects of this research was establishing a temporal strategy to predict future shorelines. Our strategy included two different variables: time depth and time horizon. Here, time depth means the number of years of shoreline data used in DSAS (e.g., 5y means five years of annual shorelines as input to DSAS for estimates of transect-specific shoreline rates). Time horizon can be defined as the year of prediction in the future. For time depth, we used three different time depths of 5, 10 and 15 years; for time horizon we used four different time horizons of 1, 5, 10 and 20 years into the future. Using 5, 10, and 15-year time depths of shoreline rates, future shoreline positions were predicted for different time horizons (e.g., 1, 5, 10, and 20 years). After deriving the predicted shoreline movement rates for each transect (e.g., LRR), we created the predicted shoreline positions along the transect using the following equation.

$$(x_t, y_t) = (((1 - t)x_0 + tx_1), ((1 - t)y_0 + ty_1)) \quad (2)$$

Where,

Start point = $(\boldsymbol{x_0}, \boldsymbol{y_0})$

End point = $(\boldsymbol{x_1}, \boldsymbol{y_1})$

Predicted Point = $(\boldsymbol{x_t}, \boldsymbol{y_t})$

The distance between Start and End point, $\boldsymbol{d} = \sqrt{(\boldsymbol{x_1} - \boldsymbol{x_0})^2 + (\boldsymbol{y_1} - \boldsymbol{y_0})^2}$

Distance ratio, $\boldsymbol{t} = \frac{\boldsymbol{d_t}}{\boldsymbol{d}}$

Predicted distance = $\boldsymbol{d_t}$

Finally, we used the geoprocessing functionality of ArcGIS Pro to create the predicted linear shorelines by connecting predicted shoreline points of sequentially adjacent transects created by the above equation.

2.6 Land Use and Land Cover (LULC) Classification and Accuracy Assessment

Based on the satellite images' spectral characteristics and familiarity with the study area's land use, we identified and classified six LULC categories for the years 1995, 1999, 2000, 2004, 2005, 2009, 2010, 2014, 2015, and 2019 (Figure 2). A detailed description of the LULC categories can be found in Table 1. We used the Random Forest (RF) and Support Vector Machine (SVM) algorithms to classify land use/cover. Random Forest is a robust machine learning classifier that balances ease of implementation and generalization ability (Rodriguez-Galiano et al., 2012). It is one of the most widely used land use land cover change classifiers (Rodriguez-Galiano et al., 2012; Zhou et al., 2020). The Support Vector Machine (SVM) algorithm is also found to be one of the most widely used algorithms in land use and land cover classification (Huang et al., 2002; Otukei and Blaschke, 2010). For a given land cover mapping task, it is often a good practice to examine multiple machine learning algorithms to compare and select a better performer (Ren et al., 2021).

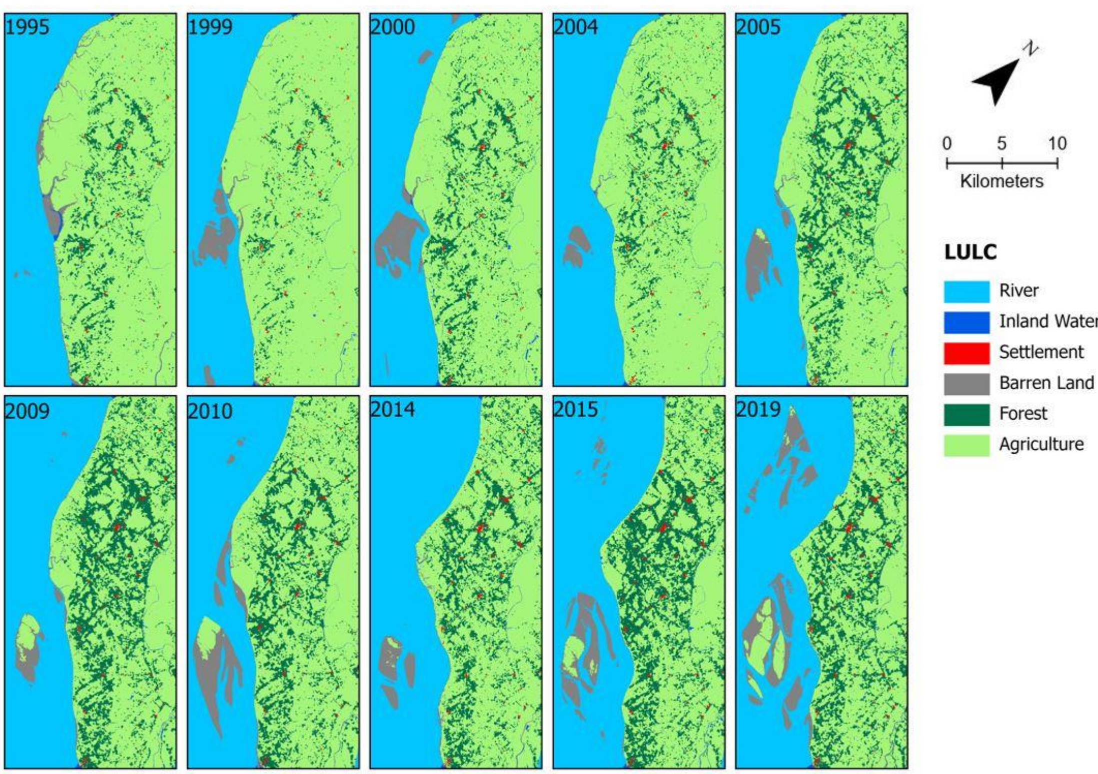

Figure 2. Land use and land cover of coastal Lakshmipur region.

Table 1. Description of the LULC categories used for the classification.

LULC category	Description
River Water	Meghna river
Inland Water	Permanent and seasonal wetlands (e.g., ponds or lakes), river tributaries, canals, and other active hydrological features
Urban/Built-up	Commercial, residential, industrial, transportation and other areas with artificial structures
Barren land	Recently developed islands, dry canals or ponds, exposed soils
Forest area	Natural or homestead forest, mixed forest lands
Agricultural land	Cultivated land, croplands, fallow lands, and vegetable fields

Post-classification accuracy is an essential part of LULC analysis. After getting the classified image, an accuracy assessment was conducted using high-resolution imagery in Google Earth Pro. Our initial assessment found that the accuracy of the SVM classifier was slightly better. As a result, we used SVM-classified LULC products for all the subsequent analyses. After getting the classified data, we found that the urban pixels are overly estimated for most of the years and underestimated for several years. To make the urban pixels consistent, we fused the Urban Settlement Footprint (WSF) data product of the German Aerospace Center (Marconcini et al., 2020). The World Settlement Footprint (WSF) evolution is a 30m resolution dataset of the global settlement extent. First, we mask out urban/built-up class using WSF results. Second, we classify the remaining study area based on our classification scheme. Finally, our classified image was fused with the WSF products using the raster calculator tool in ArcGIS Pro.

We assessed classification accuracy for the year 2015, first using the Create Accuracy Assessment Points tool in ArcGIS Pro to create 300 random points. Each land cover/use covers 60 random points. Accuracy was assessed using standard methods, including user's accuracy, producer's accuracy, overall classification accuracy, and the kappa coefficient. Our accuracy assessment result suggested that the 2015 LULC map had an overall accuracy of 86.67% and the Kappa statistic was 0.83 (Table 2). A high Kappa statistic value indicates that the relation between the predicted and actual land use is very strong. The user accuracy analysis suggested that all classes except built-up (75%) had greater than 80% accuracy. For producer accuracy, the barren and forest class had highest (96%) and lowest (75.68%) accuracy, respectively.

Table 2. Result of LULC accuracy assessment for the year 2015

LULC category	Water	Urban	Barren	Forest	Agriculture	Total	UA (%)
Water	58	0	1	0	1	60	96.67
Urban	1	45	1	12	1	60	75.00
Barren	8	0	48	1	3	60	80.00
Forest	0	1	0	56	3	60	93.33
Agriculture	0	2	0	5	53	60	88.33
Total	67	48	50	74	61	300	
PA (%)	86.57	93.75	96.00	75.68	86.89		
OA (%)	86.67						
K	0.83						

Note: UA = User's Accuracy; PA = Producer's Accuracy; OA = Overall Accuracy; K = Kappa Coefficient

2.7 Impacts of Shoreline Change on Land Use and Land Cover Calculation

Historical and predicted shorelines were overlaid with classified LULC to assess the impacts of coastal erosion on different land use and land cover. This provided historical estimation of LULC that has been impacted by coastal erosion, as well as the estimation of the future impacts. Using the actual and predicted shoreline, we created polygon to estimate the area lost. We used clip raster tool to extract area that is lost by overlaying the LULC and polygon layer. This allowed us to estimate both the total land loss and the land loss by LULC type. We used similar approach to estimate both the actual and predicted land loss for all the shoreline prediction scenarios. One goal of this overlay analysis was to quantify how well previously generated shoreline predictions perform in predicting lost or gained LULC in terms of lost area by LULC type. For example, for 2010 and 2015, we had the actual 2010 and 2015 shorelines and LULC for both years. Additionally, we produced predicted shorelines for 2015 based on the actual 2010 shoreline and using selected time depths of historical shorelines leading up to 2010 (e.g., 5yr, 10yr, 15yr time depths). These results allowed for a comparison of the actual vs. predicted lost LULC due to erosion after the 2010 to 2015 study period. A similar approach was used for other years of assessment. A detailed description of the methodology followed for predicting future shorelines using different time depths and time horizons can be found in our previously published research (Islam and Crawford, 2022). Finally, the zonal statistics tool in ArcGIS Pro was used to extract the actual and predicted land use and land cover lost.

2.8 Land loss accuracy assessment

We used recall, precision, and F1 Score metrics to evaluate the predicted land use/cover lost accuracy. The recall, precision and F1 score are confusion matrix-based metrics. The F1 score metric is a more balanced metric than recall and precision (Chicco and Jurman, 2020). All these metrics are broadly used for assessing prediction accuracy (Tseng et al., 2022). The values of the metrics mentioned above range from 0 to 1. A higher value of a metric indicates higher prediction accuracy. The metrics were calculated using the following equations. In this case, true positive (TP) means overlapping areas of actual and predicted land loss. False positive (FP) and false negative (FN) means over-predicted and under-predicted land loss areas. Figure 3. depicts a schematic diagram of land loss accuracy assessment.

$$\boldsymbol{Recall} = \frac{\boldsymbol{Ture\ Positive\ (TP)}}{\boldsymbol{True\ Positive\ (TP)\ +\ False\ Negative\ (FN)}} \qquad (3)$$

$$\boldsymbol{Precision} = \frac{\boldsymbol{True\ Positive\ (TP)}}{\boldsymbol{True\ Positive\ (TP)\ +\ False\ Positive\ (FP)}} \qquad (4)$$

$$\boldsymbol{F1\ score} = \boldsymbol{2} * \frac{\boldsymbol{Precision\ *\ Recall}}{\boldsymbol{Precision\ +\ Recall}} \qquad (5)$$

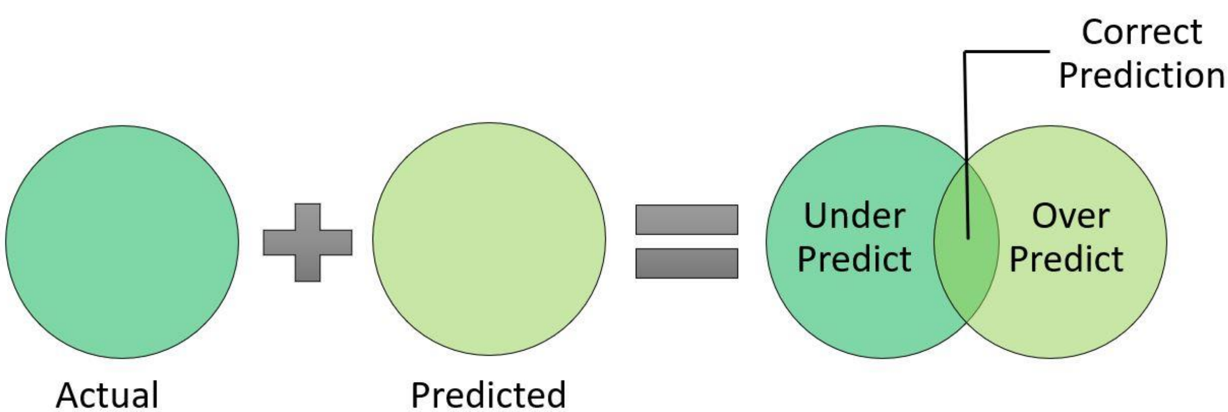

Figure 3. Schematic diagram of land loss accuracy assessment.

2.9 Statistical analysis

We used two non-parametric statistical Friedman test and Wilcoxon signed-rank test to assess the difference among models. The Friedman test is widely used for multiple comparisons of different models (Khosravi et al., 2018). We used a p-value less than 0.05 (α = 0.05) to be statistically significant. In addition to the Friedman test, we used the Wilcoxon signed-rank test since Friedman does not provide a pairwise comparison of the models. The Wilcoxon signed-rank tests are typically used to check the statistical significance of pairwise comparisons of the models (Nhu et al., 2020). The probability of the hypothesis is assessed here: whether to accept or reject a null hypothesis. The null hypothesis is rejected if there is a significant difference between different models (Nickerson, 2000).

3. Results

3.1 Spatio-temporal Patterns of Land Loss

The land loss due to erosion between 1995 – 2020 is shown in Figure 4. During the study period, agricultural land represented the LULC incurring the most loss (76.04%). This was expected, as agricultural land is the region's dominant land use/cover (Figure 2). Forest land accounted for approximately 12.93% of the total land loss. Other land cover types included barren, inland waterbodies, and settlements that were estimated to be lost at levels of 9.57%, 1.27%, and 0.19%, respectively. Previous studies suggest that most people living on this part of the coast depend on farming/agriculture and fishing for their livelihood (Paul et al., 2020; Rahman et al., 2021). As such, agricultural land loss forces people to migrate and threatens their livelihood. Among the different studied periods, the highest (3137 ha) and lowest (1367 ha) land losses were during 2010 – 2015 and 2000 – 2005. During 1995 – 2000, approximately 2285.64 hectares of land were lost. Land loss decreased in the following half-decade period (2000 – 2005). After that, it increased until the mid of the last decade. The latter half of the last decade experienced a decreased amount of land loss compared to the first half. Factors like upstream water discharge, wave energy, storm events, and heavy rainfall are possible drivers of these trends.

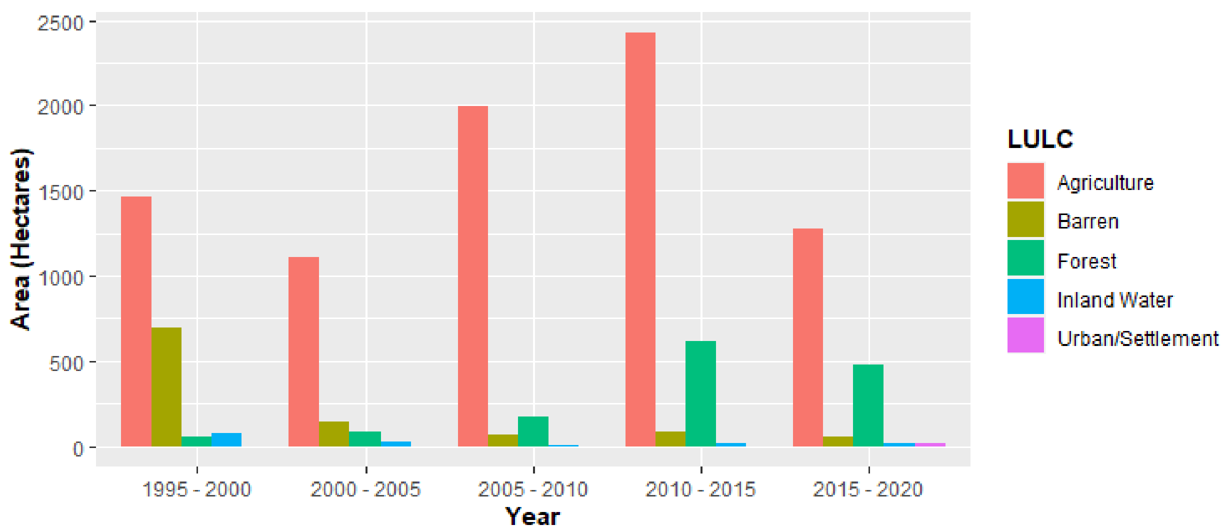

Figure 4. Historical land/LULC loss from 1995 – 2020.

3.2 Model Comparison - Total Land Loss

Land loss prediction performance was evaluated by averaging the ranking of each metric. We followed a similar ranking approach to Moayedi et al. (2020) who evaluated the performance of different image classification algorithms. First, we ranked individual metrics (e.g., recall, precision, F1) based on the prediction performance. Second, we averaged the ranking of all metrics to get the final rank. The results of individual metric ranking indicate that the best-performing model for recall was 5 year time depth and 1 year time horizon; for precision it was 15 year time depth and 10 year time horizon, and for F1, 10 year time depth and 20 year time horizon (Table 3). Our final ranking system suggests that the best model was the 10 year time depth and 20 year time horizon, followed by the 5 year time depth and 20 year time horizon model (Table 3). Our lowest-ranked model was the 15 year time depth and 1 year time horizon model. This indicates the performance was best while 10 years of shoreline data were used to predict shoreline positions 20 years in the future. This result also aligns with our previous research (Islam and Crawford, 2022), where we assessed the prediction performance of shoreline movement.

Table 3. Ranks of different combinations of time depth and time horizon evaluation for total land loss

Time Depth	Time Horizon	Count	Recall		Precision		F1		Total Rank	Final Rank
			Mean	Rank	Mean	Rank	Mean	Rank		
5	1	10	0.688	1	0.643	8	0.654	5	14	4
5	5	9	0.617	5	0.627	9	0.591	9	23	8
5	10	7	0.594	8	0.690	5	0.617	7	20	7
5	20	3	0.654	3	0.713	4	0.663	4	11	2
10	1	9	0.617	5	0.597	10	0.586	10	25	9
10	5	8	0.645	4	0.658	7	0.624	6	17	6
10	10	6	0.604	7	0.766	3	0.669	2	12	3
10	20	2	0.680	2	0.812	2	0.740	1	5	1
15	1	7	0.592	9	0.594	11	0.575	11	31	11
15	5	6	0.558	11	0.679	6	0.597	8	25	9
15	10	4	0.565	10	0.817	1	0.668	3	14	4

In addition to the ranking system, the model performance was also evaluated by the non-parametric Friedman test and Wilcoxon signed-rank test. As mentioned earlier, the null hypothesis is rejected if the p-value is less than 0.05 (α=0.05). If the p-value is less than 0.05, there is a statistically significant difference in model performance. The Friedman test results suggested a statistically significant difference in model performance for multiple models used. The lowest p-value of Friedman test was found to be 0.042, while difference in model performance was assessed using precision values for 10-year time depth and 1-year time horizon, 10-year time depth 5-year time horizon, and 10-years time depth and 10-year time horizon. The results of the Friedman's test are shown in Table 4.

Table 4. Ranking of the different models using Friedman's test.

Models	Recall		Precision		F1	
	Mean Rank	p-Value	Mean Rank	p-Value	Mean Rank	p-Value
5TD_1TH	2.29		1.43		2.43	
5TD_5TH	1.43	0.180	2.14	0.156	1.57	0.276
5TD_10TH	2.29		2.43		2.00	
10TD_1TH	1.67		1.17		1.33	

10TD_5TH	2.17	0.607	2.33	**0.042***	2.00	0.069
10TD_10TH	2.17		2.50		2.67	
15TD_1TH	2.00		1.00		1.50	
15TD_5TH	1.75	0.779	2.25	**0.039***	1.50	**0.050***
15TD_10TH	2.25		2.75		3.00	

TD = Time Depth, TH = Time Horizon, *indicates P < 0.05

The Friedman test does not provide any information on pairwise comparison. As a result, we used Wilcoxon signed rank test to discover the statistically significant performance difference between two models. The Wilcoxon test results suggest that multiple pairs of models exhibit statistically significant differences in prediction performance (α=0.05). Among different model pairs, 5-year time depth and 1-year time horizon versus 5-year time depth and 5-year time horizon exhibit a p-value less than 0.05 for precision and F1 score. Other significant pairs of models are 10-year time depth and 1-year time horizon versus 10-year time depth and 10-year time horizon with a p-value less than 0.05 (α=0.05) for both precision and F1 score. These statistically significant results of different pairs illustrate that there are multiple models where the results are significantly different from one another. Detailed results of Wilcoxon signed-rank tests can be found in Table 5.

Table 5. Performance of different models by Wilcoxon signed-rank test.

No.	Pairwise Comparison	Recall		Precision		F1	
		z-Value	P-Value	z-Value	P-Value	z-Value	P-Value
1	5TD1TH vs 5TD5TH	-1.836	0.066	-0.178	0.859	-2.310	**0.021***
2	10TD1TH vs 10TD5TH	-0.280	0.779	-0.980	.327	-0.280	0.779
3	15TD1TH vs 15TD5TH	-0.405	0.686	-1.483	0.138	-0.405	0.686
4	5TD5TH vs 5TD10TH	-1.014	0.310	-0.676	0.499	-0.845	0.398
5	10TD5TH vs 10TD10TH	-0.314	0.753	-1.153	0.249	-1.153	0.249
6	15TD5TH vs 15TD10TH	-1.461	0.144	-1.826	0.273	-1.826	0.068
7	5TD1TH vs 5TD10TH	-0.507	0.612	-0.676	0.499	-0.676	0.499
8	10TD1TH vs 10TD10TH	-0.524	0.600	-1.992	**0.046***	-2.201	**0.028***
9	15TD1TH vs 15TD10TH	-0.365	0.715	-1.826	0.068	-1.826	0.068

TD = Time Depth, TH = Time Horizon, *indicates P < 0.05

3.3 Model Comparison – Land Loss by Different LULC

Our results suggest that the model with 10 years time depth and 20 years time horizon performed the best for agricultural land, forest, and water body. The 15 years time depth and 5 years time horizon model performed the best for the urban class. In case of barren land loss prediction, the 10 years time depth and 10 years time horizon model performed the best. The 15 years time depth and 1 year time horizon model performed the worst for land cover classes of agricultural land and urban. For forest, waterbody and barren land, the worst performing models were found to be 15 years time depth and 5 years time horizon, 10 years time depth and 1 year time horizon, and 5 years time depth and 1 year time horizon model, respectively. Table 6 shows the ranking of land loss prediction performance by different LULC.

Table 6. Rank of different combinations of time depths and time horizons evaluation for land loss by LULC type.

Time Depth	Time Horizon	Final Rank				
		Agricultural Land	Forested Land	Urban /Settlement	Waterbody	Barren Land
5	1	5	3	5	10	11
5	5	8	9	3	8	7
5	10	7	7	4	5	5
5	20	4	2	10	3	9
10	1	8	7	7	11	10
10	5	6	5	2	7	5
10	10	3	4	6	2	1
10	20	1	1	9	1	3
15	1	11	10	11	9	7
15	5	8	11	1	6	4
15	10	2	6	7	3	2

While evaluating results for individual metrics used for assessing prediction performance, we found that the F1 rank had the most influence on the final ranking. It is expected because among the metrics used the F1-score was found to be most balanced metric for accuracy assessment. Due to the nature of variables used in the accuracy metric equation, the ranking result varied. For instance, in case of agricultural land, the 5 years time depth and 1 year time horizon performed best for recall. On the other hand, the best performing models based on precision and F1 score are: 15-year time depth and 10-year time horizon and 10-year time depth and 20-year time horizon models, respectively. In case of forest land, we found that 5 years time depth and 1 year time horizon model performed best based on recall value, 15-year time depth and 10-year time horizon performed best for precision and 10-year time depth and 20-year time horizon model performed best based on F1 score ranking. A detailed ranking for different metrics used can be found in Table S1, Table S2, Table S3, Table S4, and Table S5 in the Appendix.

4. Discussion

In this study, land loss due to historical and predicted shoreline movement was quantified and assessed using Landsat satellite imagery from 1988 – 2021. This study used 34 years of time series data, enabling us to assess the land loss prediction performance for a rich time series of data. The study area is located in coastal Bangladesh along the eastern bank of Meghna river. Our previous studies found that this area is highly prone to erosion (Crawford et al., 2021; Islam and Crawford, 2022). This side of the riverbank is eroding at an alarming rate; thousands of economically marginalized inhabitants of the delta are continually displaced.

The study demonstrated that overall land loss decreased from the last half of the 1990s to the mid of 2000s (Figure 5). After 2005, the land loss increased until 2015, and afterward, it decreased again. The performance of 10-year time depth and 20-year time horizon model was found to be the best (rank) for predicting future land loss. Among the different land use/cover predictions, the 10-year time depth and 20-year time horizon model was found to be the best model for predicting agricultural, forest, and inland waterbodies loss. In the urban and barren land loss prediction, the 15-year time depth and 5-year time horizon and 10-year time depth and 10-year time horizon models were found to be best performing models, respectively. Our analysis of Friedman and Wilcoxon signed-rank tests results suggests a statistically significant difference among the different model performances.

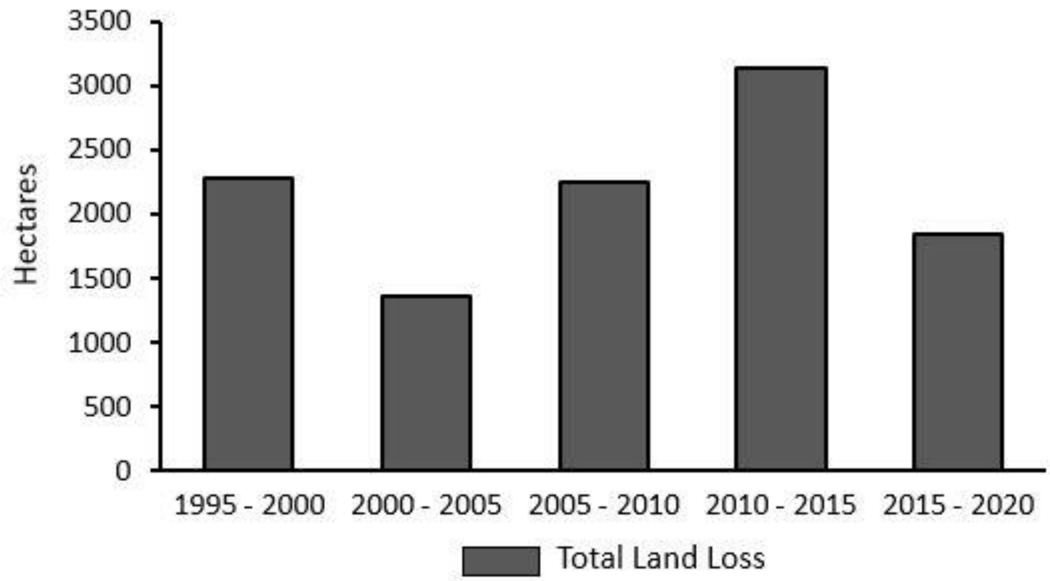

Figure 5. Total land loss in the studied area during 1995 – 2020.

Compared to much of the coastal erosion literature, this work draws from a 34 years time series of satellite-derived shorelines at annual temporal resolution. This time depth enables us to employ a temporal design strategy expected to yield a robust characterization of space-time erosion patterns. It also enables us to empirically assess the performance of future land use/cover loss predictions by predicting future shoreline positions depending on the time depths of input historical shoreline data. This is a methodological innovation with the potential for prediction of future land use/cover loss, and is applicable to other deltas and vulnerable coastal areas globally. With the increasing availability of high spatial/temporal remote sensing data, our methods can be generalized to derive a robust database of shorelines. We advocate developing and calibrating shoreline predictions using longer time-series data. While empirical results are specific to the project's study area, results can inform the region's future land use/cover loss and associated mitigation and adaptation strategies.

4.1 Limitations of the study

Our analysis provides one of the first assessments of how well predicted shoreline movement predicts amounts of LULC resources lost to erosion. Nevertheless, this study has limitations that point to future research opportunities. First, we predicted future shoreline positions solely based on the extrapolation of historical erosion rates. We know that coastal erosion rates might differ over time due to factors including flooding, cyclones, sea level rise, sediment transportation, extreme rainfall events, wave intensity, height etc. (Sanuy and Jiménez, 2019; Bamunawala et al., 2021). Since this study didn't consider these factors while predicting future shoreline positions, our prediction might get under or over-estimated for some years. Second, our analysis of LULC suggests that several classes were under or overestimated for some years. For example, urban classes were underestimated, primarily due to the built structure of these areas. Most of the houses in this part of the coast are located near/under homestead forest. As a result, we believe certain residential areas were classified as forest cover. Similar to the urban areas, water bodies like small ponds or narrow canals were classified as forest cover, and sometime even as barren land. On the other hand, we believe that the barren lands were overestimated for some years due to the depths of some wetlands or ponds being very shallow. Third, our study site is located in coastal Bangladesh where the rates of erosion are among the highest in the world. The prediction accuracy of future shoreline locations might vary based on the type, rates or location of the coast. Finally, we used Landsat imagery for both LULC and coastal erosion analysis. The pixel size of Landsat imagery is 30m. High-resolution imagery might better and accurately detect shoreline movement and LULC. Where possible, future studies might be prioritized to investigate different types of coastal landscapes with differing erosion rates.

5. Conclusion

Coastal areas are impacted by erosion globally. The severity and magnitude of coastal erosion is often high in the deltaic environments. Due to erosion, people lose their households, agricultural lands, and livelihoods. In this study, we assess the impacts of historical and predicted shoreline change on different land use and land covers in Bangladesh's Lower Meghna estuary region. This area is highly susceptible to the damaging coastal erosion. Long-term time series data at an annual scale during 1988 - 2021 were used to assess the prediction performance of different land use/cover losses. We employed a temporal design strategy of three different time depths and four different time horizons to predict future land loss for different scenarios. Agricultural lands were the top most lost land use class (76.04% of total land loss). Our results suggest that the best-performing model for land loss prediction was 10-year time depth and 20-year time horizon (rank 1) followed by the 5-year time depth and 20-year time horizon model. Among the individual land cover loss prediction, 5-year time depth and 20-year time horizon model were best performing model for the agricultural, forested and inland waterbodies loss prediction. Though the results are specific to our study area located in coastal Bangladesh, they can inform and help refine other models to predict land loss due to erosion in other parts of the coastal and deltaic environment worldwide. The results of this study provide spatiotemporally specific evidence useful for targeted coastal land management by planners, policymakers, and other relevant stakeholders.

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Appendix

Table S1. Rank of different combinations of time depth and time horizon evaluation for agricultural land loss

Time Depth	Time Horizon	Count	Recall Mean	Recall Rank	Precision Mean	Precision Rank	F1 Mean	F1 Rank	Total Rank	Final Rank
5	1	10	0.709	1	0.620	9	0.636	5	15	5
5	5	9	0.618	7	0.627	8	0.591	9	24	8
5	10	7	0.596	9	0.700	5	0.622	7	21	7
5	20	3	0.640	5	0.732	4	0.660	4	13	4
10	1	9	0.641	4	0.561	10	0.568	10	24	8
10	5	8	0.659	3	0.658	7	0.632	6	16	6
10	10	6	0.619	6	0.788	3	0.688	3	12	3
10	20	2	0.677	2	0.841	2	0.750	1	5	1
15	1	7	0.580	11	0.558	11	0.545	11	33	11
15	5	6	0.583	10	0.693	6	0.619	8	24	8
15	10	4	0.599	8	0.858	1	0.705	2	11	2

Table S2. Rank of different combinations of time depth and time horizon evaluation for forest loss

Time Depth	Time Horizon	Count	Recall Mean	Recall Rank	Precision Mean	Precision Rank	F1 Mean	F1 Rank	Total Rank	Final Rank
5	1	9	0.724	1	0.455	8	0.514	3	12	3
5	5	9	0.524	4	0.436	9	0.426	8	21	9
5	10	7	0.489	7	0.467	7	0.438	6	20	7
5	20	3	0.640	2	0.486	5	0.531	2	9	2
10	1	8	0.500	5	0.483	6	0.399	9	20	7
10	5	8	0.477	8	0.513	4	0.442	5	17	5
10	10	6	0.489	6	0.530	3	0.478	4	13	4
10	20	2	0.621	3	0.534	2	0.575	1	6	1
15	1	6	0.442	9	0.424	11	0.366	10	30	10
15	5	6	0.313	11	0.435	10	0.337	11	32	11
15	10	4	0.355	10	0.549	1	0.430	7	18	6

Table S3. Rank of different combinations of time depth and time horizon evaluation for settlement loss.

		Count	Recall	Precision	F1	Total

Time Depth	Time Horizon		Mean	Rank	Mean	Rank	Mean	Rank	Rank	Final Rank
5	1	2	0.865	2	0.345	7	0.433	5	14	5
5	5	2	0.931	1	0.492	5	0.650	3	9	3
5	10	2	0.606	5	0.565	2	0.585	4	11	4
5	20	2	0.341	8	0.144	10	0.202	10	28	10
10	1	2	0.450	6	0.158	9	0.223	9	24	7
10	5	2	0.854	3	0.534	3	0.657	2	8	2
10	10	2	0.302	10	0.529	4	0.364	6	20	6
10	20	2	0.314	9	0.189	8	0.236	8	25	9
15	1	2	0.412	7	0.135	11	0.199	11	29	11
15	5	2	0.760	4	0.613	1	0.678	1	6	1
15	10	2	0.177	11	0.370	6	0.238	7	24	7

Table S4. Rank of different combinations of time depth and time horizon evaluation for inland waterbody loss

Time Depth	Time Horizon	Count	Recall		Precision		F1		Total Rank	Final Rank
			Mean	Rank	Mean	Rank	Mean	Rank		
5	1	10	0.690	10	0.702	11	0.688	10	31	10
5	5	9	0.713	9	0.796	7	0.745	8	24	8
5	10	7	0.800	7	0.868	2	0.823	5	14	5
5	20	3	0.803	6	0.877	1	0.835	4	11	3
10	1	9	0.649	11	0.728	10	0.670	11	32	11
10	5	8	0.810	5	0.791	8	0.798	7	20	7
10	10	6	0.855	2	0.867	3	0.857	2	7	2
10	20	2	0.876	1	0.858	4	0.867	1	6	1
15	1	7	0.728	8	0.773	9	0.743	9	26	9
15	5	6	0.832	4	0.798	6	0.812	6	16	6
15	10	4	0.840	3	0.849	5	0.838	3	11	3

Table S5. Rank of different combinations of time depth and time horizon evaluation for barren land loss

Time Depth	Time Horizon	Count	Recall		Precision		F1		Total Rank	Final Rank
			Mean	Rank	Mean	Rank	Mean	Rank		
5	1	10	0.713	11	0.839	11	0.759	11	33	11
5	5	9	0.751	8	0.943	8	0.824	8	24	7
5	10	7	0.835	6	0.966	4	0.888	6	16	5
5	20	3	0.739	9	0.959	7	0.822	9	25	9
10	1	9	0.729	10	0.847	9	0.772	10	29	10
10	5	8	0.854	5	0.962	6	0.901	5	16	5
10	10	6	0.908	1	0.987	3	0.945	1	5	1

10	20	2	0.880	3	1.000	1	0.936	3	7	3
15	1	7	0.814	7	0.843	10	0.826	7	24	7
15	5	6	0.870	4	0.965	5	0.912	4	13	4
15	10	4	0.897	2	0.993	2	0.942	2	6	2

Chapter 4. Erosion experience, geophysical vulnerability, and coastal erosion risk perceptions in Bangladesh

This chapter assesses the patterns and drivers of erosion risk perception using data from spatially explicit measures of coastal erosion risk derived from satellite imagery and a random sample survey of residents living in the coastal communities. It is currently being revised for submission to the journal "*Journal of Environmental Management*".

Islam, MS., Crawford, TW., Juran, L., and Higdon, DM. (2023). Erosion Experience, Geophysical Vulnerability, and Coastal Erosion Risk Perceptions in Bangladesh. Target Journal: "*Journal of Environmental Management*".

Abstract

Coastal Bangladesh has one of the highest rates of coastal shoreline erosion in the world. Erosion causes numerous problems including population displacement and loss of houses and household assets, productive lands, infrastructure, and livelihoods. Due to climate change and associated sea level rise, it is expected that erosion intensity and magnitude will increase in the future. This study aims to advance the understanding of the drivers of erosion risk perception using spatially explicit measures of coastal erosion risk derived from satellite imagery and a random survey of 407 residents of the eastern bank of the Meghna Estuary. Prior empirical work has documented that the Meghna Estuary in coastal Bangladesh has experienced extreme rates of erosion since 2000. With an aim to assess the drivers of risk perceptions, this study uses Logistic Regression (LR) modeling to examine the roles of demographic, economic and location variables as predictors of accurate or inaccurate perceptions of subsequently measured erosion occurrence.

Results suggest that the accurate prediction of erosion risk by surveyed households is significantly influenced by the physical location of the house. Further, predictors such as proximity to the coast and whether respondents' house locations were unprotected by a recently constructed revetment are strongly associated with the accurate prediction of coastal erosion risk. This study highlights the critical role socio-economic and locational attributes play in risk perception at the household scale. Findings of this study inform the need to raise awareness for better planning and management of coastal areas and thus resilience of coastal populations to a changing climate. Research findings can also assist in the development of associated mitigation and adaptation measures that better incorporate community perceptions of risk.

Keywords: Coastal Erosion; Vulnerability; Risk Perception; Coastal Communities; Meghna River; Bangladesh

1. Introduction

Coastal erosion is one of the major environmental threats globally (Parry et al. 2007; Williams et al. 2018). In Bangladesh, it is one of the major natural disasters that occurs regularly (Ahmed et al. 2018; Crawford et al. 2021; Sarwar and Woodroffe 2013). The lower Meghna estuary has one of the highest erosion rates in the world and is thus one of the most vulnerable areas to coastal erosion (Crawford et al. 2021). Rapid geomorphological changes are occurring in the region due to erosion; however, new land is also being created due to accretion of deposited sediments from erosional sites. According to Brammer (2014), the Meghna estuary had a net land gain of 451 km^2 from 1984 to 2007 with an average growth rate of 19.6 km^2 annually.

Erosion results in a loss of productive land area, loss of infrastructure and communication systems, population displacement, and most importantly impacts household livelihoods (Bhuiyan et al. 2017; Crawford et al. 2020). Due to extensive erosion in the Meghna estuary, thousands of residents are displaced annually. As a low-income country with limited resources, Bangladesh struggles to cope with coastal erosion--which acutely impacts rural, coastal communities through unemployment and loss of livelihood. As a result, socio-economic conditions of the displaced people are deteriorated. Most of the displaced become landless, which pushes them into further forced migration and the poverty cycle.

Climate change projections are adding extra risk for people living near coastlines. Although the response of world's coastlines will vary by region, it is predicted that poorer countries will suffer more. A study on the distributional impact of climate change on relatively wealthy and poor countries by Mendelsohn et al. (2006) found that although wealth, adaptation and technology influence distributional consequences, across countries the primary reason why relatively poor countries are so vulnerable is their geographic location. Meanwhile, people living in the same geographic area will be disproportionately impacted by the climate crisis due to various social vulnerabilities. Thus, among an affected population, poorer cohorts may be more heavily impacted. A report from the Demographic and Health Survey (DHS) conducted by the World Bank in multiple countries suggests that the relatively poor lose assets two to three times more than non-poor people when affected by a natural disaster (Park et al. 2015; Hallegatte and Rozenberg 2017).

Understanding risk perceptions is critical for improving risk communication because it offers insight on the drivers of risk perception and thus equips risk communicators to better manage and plan for natural hazards and disasters. While drivers of risk perception vary according to the system, Carlton and Jacobson (2013) contend that several factors represent critical components including social trust, risk salience, environmental attitudes, and demographic factors. Individuals with higher social trust may perceive less risk compared to those with less social trust (Bronfman et al. 2016). In terms of environmental attitudes, Slimak and Dietz (2006) found that higher levels of environmental concern are related to higher levels of risk perception. Risk salience also influences risk perception. Generally, risk salience has two components: relevant prior experience and proximity to the risk (Carlton and Jacobson 2013). Further, proximity to the coast had a substantial influence on perceived risk as individuals residing near the coast were more exposed to natural hazard events compared to individuals living inland. Lastly, risk perceptions are influenced by demographic factors. Many studies have found that factors such as gender, political affiliation, marginalization and educational background influence perceptions of risk (Slovic 1999; Linden et al. 2015; Sund et al. 2017). A study on socio-physiological determinants of climate change risk perception by Van der Linden (2015) found that gender, political party, knowledge of the causes, social norms, value orientations, impacts and responses to climate change, personal experience and affect with extreme weather were significant predictors. Furthermore, it has been established that living in a historically hazardous place can influence risk perception (Lujala et al. 2015).

This research aims to understand the patterns and drivers of human perceptions of coastal erosion risk. The study area is located along the eastern bank of the Lower Meghna river in Bangladesh. To evaluate what drives individuals risk perceptions, we combine spatially explicit measures of coastal erosion risk (e.g., actual erosion rates derived geospatially) and human perceptions of erosion risk to investigate the accuracy of coastal erosion risk prediction. Using multitemporal satellite imagery and household survey data, we seek to understand the status and drivers of coastal erosion risk perception in the region. A significant feature of this study is the temporal sequencing of primary social survey data vis-à-vis historical measurements of shoreline retreat causing actual surveyed locations to be lost into the Meghna estuary. The social survey data are described below in more detail. These data were collected in May-June 2018 and included questions on risk perception and respondent predictions regarding if and when their home site would be lost to erosion. Post-2018, erosion caused significant shoreline retreat enabling us to measure if respondents in 2018 accurately predicted if and when their homes would be lost. We hypothesized that predictors such as education level, locational attributes and other factors are strongly correlated with resident's accurate prediction of coastal erosion risk. The specific objectives of this study were to: (1) assess the number of household sites lost to erosion during 2018-2022, (2) determine the status of coastal erosion risk perceptions among the coastal communities, (3) examine the correlation of age, gender, income, length in residence, prior erosion displacement experience, location, and other factors with the accuracy of coastal erosion risk prediction, and (4) evaluate what drives the respondents' accurate prediction of future erosion risk.

2. Materials and Methods

2.1 Study Area

The study area is located on the eastern bank of Meghna estuary (Figure 1) near the Bay of Bengal--part of one of the biggest deltas in the world. The study area covers Ramgati upazila of Lakshmipur district in Bangladesh (upazilas are the second lowest tier of administrative unit). The land use and land cover of this region is dominated by agriculture (Islam et al. 2023), and most households in the area coast depend on agriculture or fishing for livelihood (Rahman et al. 2021; Paul et al. 2020). The Lower Meghna estuary is highly vulnerable to coastal erosion (Islam et al. 2022; Crawford et al. 2021). Residents living in the region are situated very close to the sea level, with an average elevation of 5m.

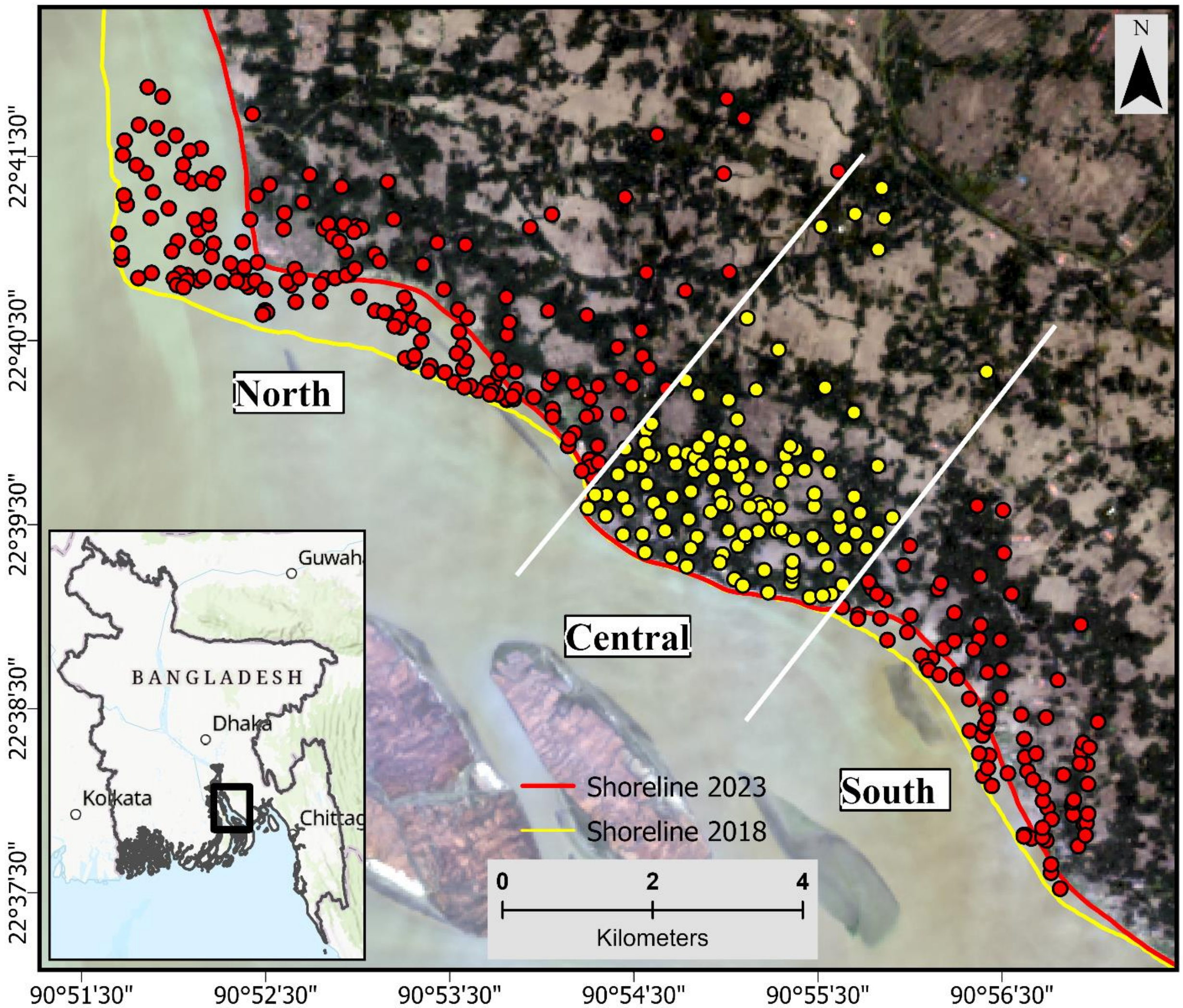

Figure 1. Study area map with survey locations. Yellow dot locations are protected by concrete embankment while the red dot locations are unprotected.

2.2 Data

This research employs multiple sources of data. Landsat satellite data were obtained from the USGS Earth Explorer website (https://earthexplorer.usgs.gov/). This dataset was used to map annual shoreline from 2008-2023. Since cloud cover is an issue in this region, we obtained imagery only from the months of November-February at approximate annual intervals to ensure cloud-free images. Landsat imagery is widely used for quantification of coastal erosion (Crawford et al. 2021; Ahmed et al. 2018). A detailed description of the imagery used in this study is provided in Table 1.

Table 1. Landsat imagery used in the study.

Image date (mm/dd/yyyy)	**Representative year**	**Satellite**	**Path/Row**	**Resolution (m)**
12/16/2007	2008	Landsat 7	137/44	30
01/03/2009	2009	Landsat 7	137/44	30
01/30/2010	2010	Landsat 5	137/44	30
01/01/2011	2011	Landsat 5	137/44	30
01/28/2012	2012	Landsat 7	137/44	30
01/14/2013	2013	Landsat 7	137/44	30
12/24/2013	2014	Landsat 8	137/44	30
11/25/2014	2015	Landsat 8	137/44	30
12/30/2015	2016	Landsat 8	137/44	30
01/01/2017	2017	Landsat 8	137/44	30
01/04/2018	2018	Landsat 8	137/44	30
01/07/2019	2019	Landsat 8	137/44	30
01/10/2020	2020	Landsat 8	137/44	30
12/27/2020	2021	Landsat 8	137/44	30
01/07/2022	2022	Landsat 9	137/44	30
01/10/2023	2023	Landsat 9	137/44	30

Original household survey data (n=407) were used for analyzing respondents' risk perception regarding coastal erosion. Survey questions were designed to measure information about respondents' sociodemographic and economic attributes as well as their perceptions of coastal erosion risk in Bangladesh. These data were collected as part of a project funded by the National Science Foundation (award #1660447). The survey covers an area along a 15km stretch of shoreline located in Ramgati upazila in Lakshmipur district. Sample locations were limited to the near shoreline area extending approximately 5km to the interior of the 2018 shoreline. Data collection followed an approved IRB protocol. Survey responses are spatially explicit, meaning that GPS latitude-longitude coordinates were obtained for each household location. Figure 2 provides a summary of respondents' sociodemographic characteristics.

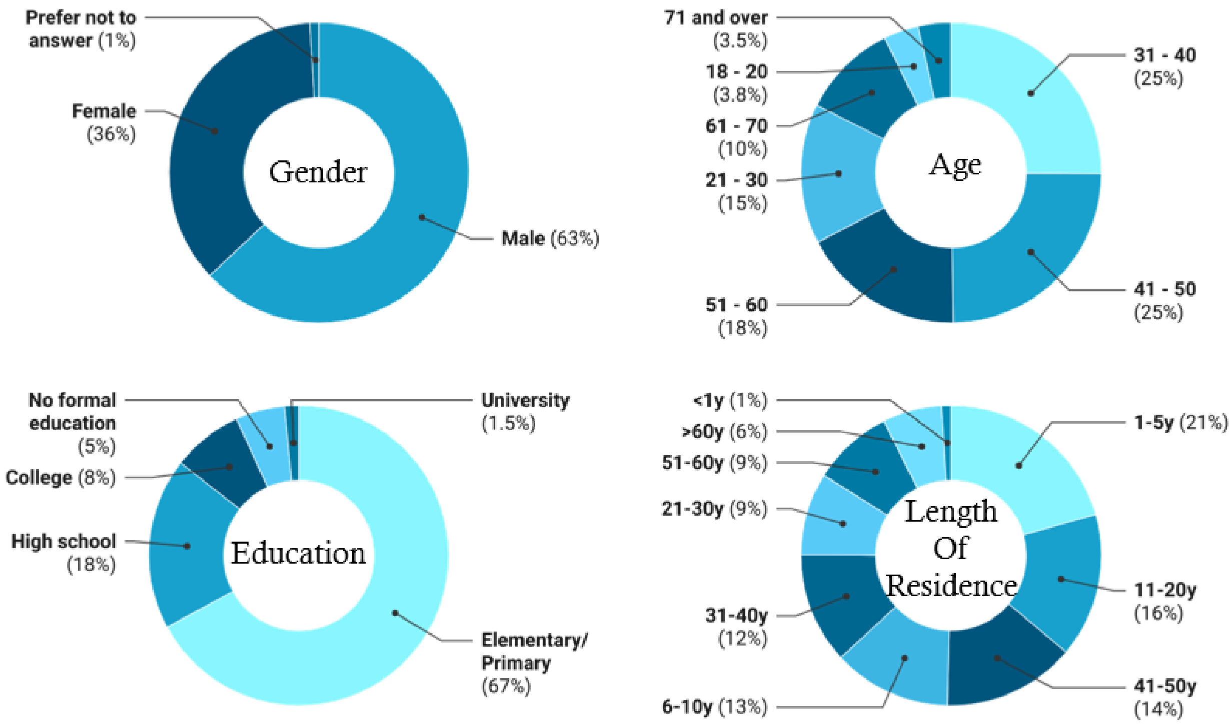

Figure 2. Sociodemographic characteristics of survey respondents.

2.3 Methodology

2.3.1 Shoreline Extraction

The shoreline extraction process followed four steps. First, we calculated the Modified Normalized Difference Water Index (MNDWI). MNDWI is defined as:

$$\mathrm{MNDWI} = \frac{\mathrm{Green-MIR}}{\mathrm{Green+MIR}} \qquad (1)$$

The middle infrared and green bands were used to calculate MNDWI index to enhance the ability to distinguish between open water and non-open water sources (Xu, 2006). The value of the index ranges from -1 to 1, where water pixels approach 1 and are easily distinguishable from other pixels. Second, we used ISO cluster unsupervised classification to classify the image into 10 classes. The Red, Green, Blue and MIR bands along with MNDWI were used as inputs. Third, the classified image was reclassified into two classes, water and non-water, via visual interpretation. Finally, various GIS processing tools and techniques were used to generate annual vector shorelines.

After extracting the shorelines, the shoreline uncertainty was assessed. Different variables were taken into consideration to assess uncertainty. A detailed description of the procedure can be found in our previous studies (Crawford et al. 2021; Islam and Crawford 2022). Shoreline uncertainty was assessed using the equation of Hapke et al., 2010:

$$Utotal = \sqrt{Ug^2 + Up^2 + Ud^2 + Ut^2} \qquad (2)$$

Where, *Utotal* = total uncertainty, *Ug* = georeferencing uncertainty, *Up* = pixel uncertainty, *Ud* = digitizing uncertainty and *Ut* = uncertainty associated with tidal variation.

2.3.2 Variables

2.3.2.1 Predictor Variables

Survey responses

The survey consisting of 95 questions was administered in the coastal Ramgati region of Lakshmipur district during the summer of 2018. Respondents were selected randomly by visiting households located near the shoreline. We collected 420 responses from individuals with at least 18 years of age. We initially cleaned the dataset based on the geographic locations of the responses. Other cleaning procedures included removing duplicates and checking for logical consistency, haphazardly given answers and nonsensical responses via inspection of the descriptive statistics and frequencies. A final sample of 407 surveys resulted after the cleaning strategy.

Out of the 95 questions, only 20 questions were utilized in this study. Most questions were multiple choice focusing on respondents' sociodemographic characteristics and perceptions of coastal erosion risk. Most of the risk perception questions included Likert-scale responses with five categories ranging from strongly disagree to strongly agree; an example question is "*Please indicate how much you agree or disagree with the statement, "Riverbank erosion has increased since 1990*". The response categories were strongly disagree =1, disagree =2, neither disagree or agree =3; agree =4, and strongly agree =5. In addition to multiple choice questions, we also collected information such as respondents' age, individual income, household income, education level, and other relevant variables.

Geospatial variables

This study developed geospatial variables using shoreline and household location data. We used survey locations (i.e., at the households) to calculate the distance between nearest shoreline and the corresponding physical houses. Using a similar approach, we calculated the distance between the houses that were surveyed and historical shorelines for years 2008, 2014, and 2018, which also enabled the measurement of GIS-derived shoreline retreat (i.e., erosion, or how much a respective shoreline has been approaching a household site). We also used satellite imagery to develop a variable that identifies the presence or absence of a protective embankment between household sites and the shoreline. A new revetment had been constructed shortly before survey data collection in 2018 that protects approximately 1.5km of the shoreline. Additionally, we used the *Buffer* tool in ArcGIS Pro to categorize (e.g., near, far) the household locations based on their proximity to the shoreline (Table 2).

Table 2. Locational attributes of survey respondents

	N	%
Revetment between HH and 2018 shoreline		
Yes	122	30%
No	285	70%
Distance between HH and 2018 shoreline		
<500m	142	35%
500 to 1500m	195	48%
>1500m	70	17%

2.3.2.2 Outcome Variable

Accuracy of perceived risk was used as the outcome variable in the logistic regression model. The accuracy of perceived risk was determined by comparing whether survey respondents' actual house was lost (or not) lost due to coastal erosion against their perceived risk on whether their house would be lost to erosion at specified dates after data collection in 2018. Dates for potential house loss were specified in discrete years (i.e., house will be lost by the end of 2018, 2019, 2020, etc.). A confusion matrix was used to determine who predicted accurately and who overestimated/underestimated (Figure 3). In the figure, TP = accurately predicted house will be lost by 2022, TN = accurately predicted that house will not be lost by 2022, FN = overpredicted that house will be lost by 2022 (i.e., predicted house would be lost but it was not), and FP = underpredicted that house will be lost by 2022 (i.e., predicted house would not be lost but it was). We combined TN and TP responses as accurate predictions, while FN responses represented inaccurate predictions. Empirically in terms of a temporal technicality, there were no FP responses.

	Actual Risk	
Predicted Risk	True Positive (TP)	False Negative (FN)
	False Positive (FP)	True Negative (TN)

Figure 3. Schematic of the confusion matrix to classify actual vs. predicted risk.

2.3.3 Statistical Analysis

First, we explored the descriptive statistics of different variables (independent and dependent) used in the study. This provided a useful summary of the different types of data to be used. Second, we performed a chi-squared test of independence to assess the correlation between the nominal variable 'prediction of coastal erosion' (i.e., accurate or inaccurate prediction) and all other ordinal and nominal variables. In some cases, two categories were combined to satisfy the chi-squared test of significance assumptions. Third, we deployed a binary Logistic Regression (LR) to understand the relationship between the accurate prediction of coastal erosion risk and other variables. The LR model has been widely used to assess perceived risk in natural hazards research (Mind'je et al. 2019; Shah et al. 2022). The correlation analysis using chi-squared tests of independence served as a method to reduce the number of variables from over 40 candidate variables used in the LR model. Further variables reduction was achieved through the Backward Selection method, resulting in a final LR model.

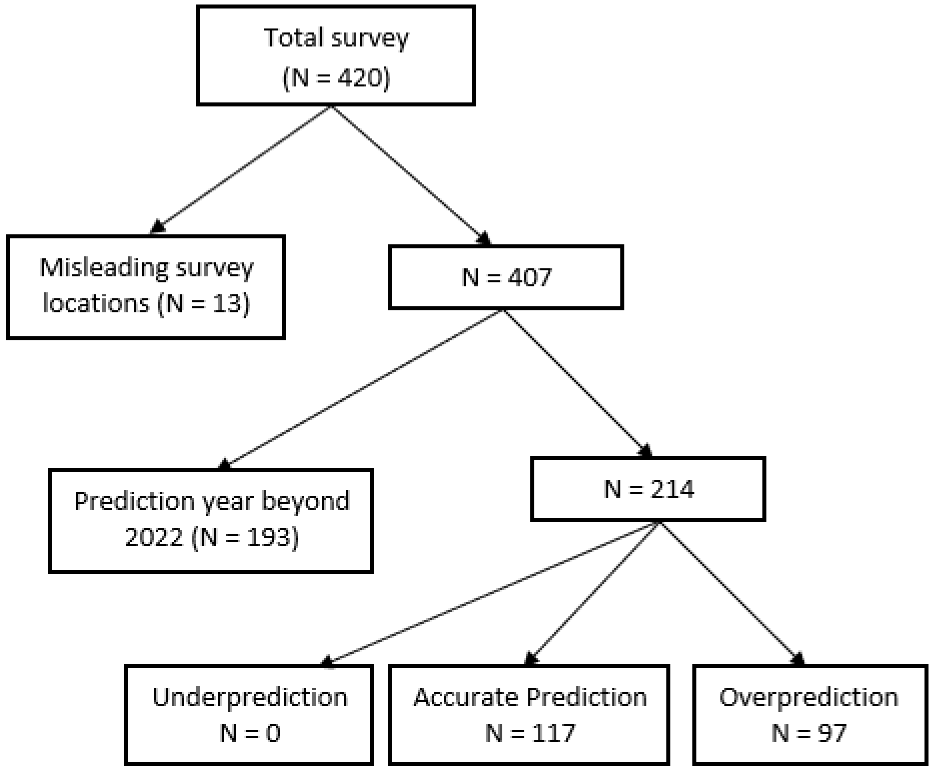

Figure 4. Description of survey selections for the logistic regression model.

Figure 4 portrays the survey selection process for the LR model. The prediction outcomes in the bottom row of Figure 4 served as inputs to the dependent variable measure of accurate/inaccurate prediction. Zero respondents underpredicted, meaning that no respondents perceived that their house would be lost by end of year 2022 when in fact it was lost. “Accurate Prediction” (n = 117) means that respondents correctly predicted that their house would or would not be lost by 2022. “Overprediction” (n = 97) means that respondents incorrectly predicted that their house would be lost by 2022 when in fact it was not lost. These two outcomes form the binary variable used as the dependent variable where Accurate Prediction = 1 and Overprediction = 0.

Predictor variables included both nominal and ordinal data. In the case of Likert responses, data were coded as follows: Strongly disagree = 1, Disagree = 2, Neutral = 3, Agree = 4 and

Strongly agree = 5. Similarly, binary responses were transformed as: No = 0 and Yes = 1. We defined a p-value less than 0.05 as statistically significant. We report both p-value and odds ratio for each predictor variable in the final regression model. An odds ratio greater than 1.0 indicates a positive association between the response and the predictor variable, while an odds ratio less than 1.0 indicates a negative association.

3 Results

3.1 Status of respondents' locations in 2023

As part of the survey, we logged the geolocation of respondents' houses. This survey data from 2018 combined with recent Landsat imagery data through 01/2023 allowed us to assess the actual magnitude of post-survey coastal erosion and thus whether household sites were actually eroded. To assess the number of households lost by erosion, we overlaid the locations of the survey/houses with the annual shoreline derived from satellite imagery (Figure 5). Our analysis found that, by the end of the 2022, 141 out of 407 survey/house locations were lost due to erosion (Table 3). This represents roughly 35% of the total survey. While assessing the magnitude of houses lost by year, approximately 11% of the houses were lost into the river in 2021 alone. An annual summary of houses lost from 2018 – 2022 is presented in Table 3.

Table 3. Summary of the number of houses lost 2018 – 2022.

Year	Number of houses lost	Percentage of houses lost	Cumulative number of houses lost	Cumulative % of houses lost
2018	20	4.9	20	4.9
2019	23	5.7	43	10.6
2020	38	9.3	81	19.9
2021	46	11.3	127	31.2
2022	14	3.4	141	34.6

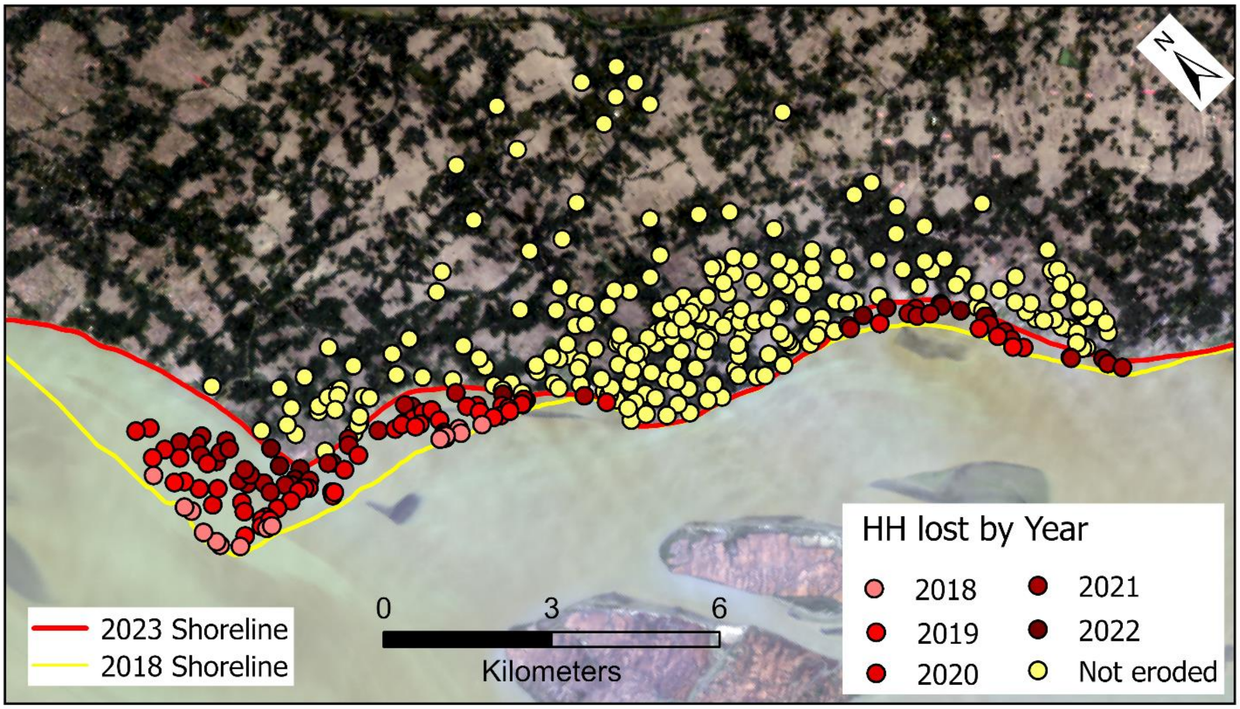

Figure 5. Survey respondent's household locations lost 2018 – 2022.

3.2 Descriptive statistics of the socioeconomics, demographics, and risk perception variables

The survey respondents were 63% male and 36% female. The majority of respondents had elementary/primary school education (67%), while roughly 9.5% had a college degree and 5% possessed no formal education. Among age groups, the majority of respondents (68%) ranged 30 – 60 while approximately 19% were 18 – 30 years. Respondents aged 60 and older comprised about 13% of survey respondents. As for the family size, all respondents had at least two family members and a majority (76%) had a family size of five or more members.

The correlation analysis suggests that the accuracy of respondent erosion risk prediction is strongly related to their education ($p = <0.001$) while being less related to their respective age ($p = 0.069$). Among the education categories, respondents with primary education (64%) were more accurate in predicting future erosion risk than respondents with greater levels of education (26%). The low percentage of accurate predictions among respondents with relatively greater educational attainment (i.e., high school or higher degrees) is attributable to the fact that the cohort was markedly 'liberal' in predicting future erosion risk. In other words, respondents with greater education tended to overpredict future coastal erosion—meaning that they predicted their house would be eroded before the end of 2022, but it was not. As for age groups, respondents aged 30-40 and 51-60 were more accurate in predicting future erosion risk (64% in 30-40 and 63% in 51-60 cohorts) compared to the other age cohorts.

The survey had multiple questions probing respondents' knowledge of historical and future coastal erosion in Bangladesh. One question asked, "if the riverbank erosion increased since 1990". Virtually all respondents (96%) agreed that erosion had increased since 1990 (Figure 6). Similar responses were found for a question related to coastal erosion. When asked "if more information about future riverbank erosion will be useful," 86% of respondents agreed that additional information on future erosion risk would be useful for enhanced planning and adaptation ($p = 0.012$). However, most respondents disagreed when asked if they receive useful information on coastal erosion from non-governmental organizations ($p = 0.464$), the Government of Bangladesh ($p = 0.004$), and local leadership ($p = 0.026$). While assessing if having information services and devices (e.g., TV, radio, internet) has any relation with the accurate prediction of future erosion risk, we found that there is a strong correlation between prediction accuracy and possessing communication devices and services such as a TV, radio, laptop, internet, etc. ($p = <0.001$).

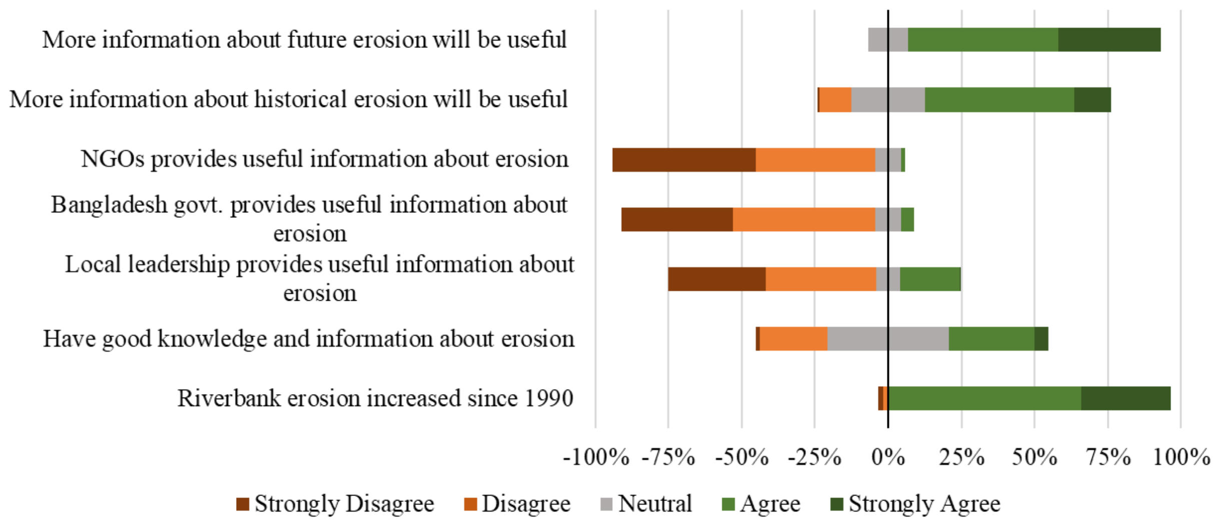

Figure 6. Distribution of Likert-scale responses measuring respondents' knowledge of coastal erosion.

An exploration of respondents' perceptions of shoreline protection strategies determined that most respondents believed that concrete embankments are needed to protect the shorelines in their neighborhoods and elsewhere. For example, one question asked if a concrete embank is needed to protect the nearby shoreline, and the overwhelming majority of respondents agreed (99%). The survey also asked whether a concrete embankment is needed to protect the entire eastern bank of Meghna River shoreline (i.e., the side of the shore that was surveyed), with 95% of respondents answering positively (Figure 7). A similar question asked if embankment construction is positive or negative. Nearly 97% of respondents reported that the construction of a concrete embankment in their neighborhood would be considered a good thing.

We also explored if there is a relation between respondents' accurate prediction of erosion risk and embankment protection. A survey question asked if there is an embankment (e.g., with large blocks, concrete, etc.) located between your home and the Meghna riverbank. The correlation analysis suggests that accurate predictions and the (perceived) existence of a constructed embankment between the respective houses and nearest riverbanks are highly correlated (p = <0.001). We also checked the correlation of embankment locations with accurate predictions by analyzing satellite imagery with embankments geolocated. The chi-squared test of independence suggests that there is a strong correlation between locations with embankment protection and respondents with accurate predictions of erosion (p = <0.001).

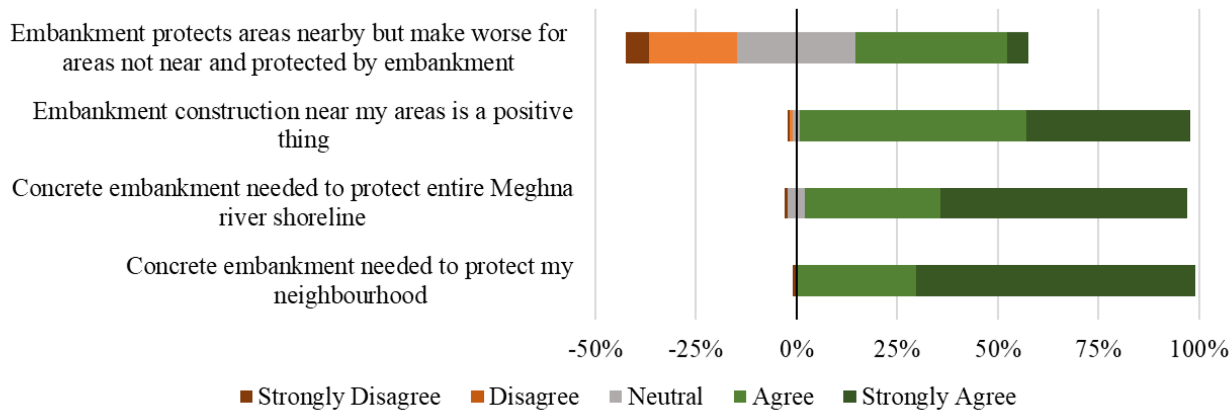

Figure 7. Distribution of Likert-scale responses measuring respondents' perceptions of concrete embankment protection.

Most survey respondents (83%) were located within 1.5 kilometers of the shoreline, with 35% located within 500 meters of the shoreline. The chi-square test determined that distance of house locations from the shoreline (irrespective of the existence of an embankment) is related to the accurate prediction of erosion risk (p = <0.001). Respondents located nearer to the shoreline exhibited higher accuracy in risk prediction compared to those located relatively farther from the shoreline. Respondents living within 500 meters of the shoreline had the highest accuracy (75%), followed by respondents living 501 – 1500 meters from the shoreline (40%) and beyond 1500 meters (0%). This indicates that people living near the shoreline are more knowledgeable of the behavior of shoreline movement. We also explored if specific geographic regions of the study area (i.e., north, central, south) were related to the accurate prediction of coastal erosion risk. We found that those living in the north region are much more accurate (67%) compared to those located in the south (42%) and central (0%) regions. A high p-value (p = <0.001) of chi-square test demonstrates the strong relation with regions and the accurate prediction of future erosion risk. This makes sense because the north region was more exposed to erosion compared to the central and south regions for being located in the upstream of the river. Meanwhile, residents in the central region are protected by a concrete embankment but expressed concern that the embankment would soon fail. As a result, respondents living in this region of the shoreline overpredicted that their locations will be eroded by 2022.

3.3 Drivers of accurate predictions of coastal erosion risk

Results of the final binary logistic regression model are presented in Table 4. Among the sociodemographic considerations, only education was found to be a negative predictor of accurate prediction of coastal erosion. When moving from primary school or less to higher school education, the odds of accurate prediction are decreased by 75.3%. Among the locational attributes, distance between respondents' houses and the shoreline and regions of the respondents were negative predictors of accurate prediction. Respondents who lived near the shoreline and in the north region were more accurate in predicting erosion compared to those living farther from the shore and in the central and south regions. As a respondent household site location moves along the scale from nearer to farther from the 2018 shoreline, the odds of accurate prediction to over prediction decrease by 89.3% at each step. In case for north to south region, the odds of accurate prediction decrease by 86.1% at each level (e.g., north, central, south).

Having an electricity connection in the household was also found to be a predictor of accurate predictions of erosion risk. An odds ratio value of 0.170 with a negative coefficient indicates that the odds of accurate prediction to over prediction decrease by 83%. A Likert-scale question inquiring if "my village, union, or upazila leadership provides useful information about the problem of riverbank erosion that I am able to use effectively" was also found to be a predictor of accurate predictions (negative coefficient). As the scale of the responses move from strongly disagree to strongly agree, the odds of accurate prediction compared to overprediction decrease by 60.8% for each step.

Table 4. Results of final binary logistic regression model showing the relationship between the response and predictor variables.

Variables	**Coef**	**P-value**	**OR**	**95% OR CI**
Constant	4.097	**<0.001**	60.183	
Completed level of education	-1.399	**0.005**	0.247	0.094, 0.650
Distance between HH and shoreline	-2.231	**<0.001**	0.107	0.043, 0.268
Regions (north, central, south)	-1.970	**<0.001**	0.139	0.052, 0.375
Electricity in the HH	-1.772	**<0.001**	0.170	0.073, 0.395
Bangladesh govt. provides useful information on riverbank erosion	-0.937	**0.036**	0.392	0.163, 0.939

OR = Odds Ratio; CI = Confidence Interval

4 Discussion

This study aimed to investigate the predictors of accuracy of coastal erosion risk. Compared to much of the existing literature that explores risk perceptions solely based on survey responses, this research utilized human perceptions of erosion risk (i.e., surveys) combined with spatially explicit measures of erosion (i.e., actual annual erosion derived geospatially). A binary logistic regression model was designed to understand how socioeconomic, demographic and locational attributes explain the variability of respondents' accuracy in future coastal erosion risk.

The results suggest that there is a strong correlation between the accuracy of erosion risk prediction and education. Though it was hypothesized that relatively educated respondents would be more accurate in predicting future erosion risk, this study found that respondents who are less educated (no formal education to primary school education) were more accurate in predicting future erosion compared to those with at least high school education. Previous experience with erosion might have an influence on why less educated individuals were more accurate in predicting future erosion risk. A study on public perceptions of risk by Wang et al. (2018) reported that respondents with higher educational backgrounds were less worried and aware of flood risk in China. Risk perception is significantly influenced by the physical location of the respondents. Our analysis also indicates that those residing closer to shoreline and those without embankment protection were more accurate in predicting future erosion risk. Similar results were also found in the region when investigating resident perceptions of riverbank erosion and shoreline protection (Rahman et al. 2022), and this coincides with Carlton and Jacobson's (2013) contention that proximity (to the risk) matters. However, a study by Ali et al. (2022) found contrasting results when examining flood risk perception in Pakistan. The authors found that people living farther from the river exhibited higher risk perception compared to those living relatively nearer.

Risk perceptions are significantly influenced by the availability and accessibility of information technologies. A majority of respondents (86%) agreed that information on future erosion risk will be useful for enhanced future planning. This study also found that the possession of information services and devices (e.g., mobile phone, TV, radio, internet) is strongly correlated with the accurate prediction of future erosion risk. Having access to such devices and services keeps people informed about what to expect during a coastal erosion event. Property loss and the disruption of daily activities can be mitigated through communication of information to residents during a disaster event (Aksha et al. 2019, Bica et al. 2020)—but proper communication is not possible if residents lack access to services and devices such as mobile phones, the internet, TV and radio.

While assessing the drivers of accurate prediction of future erosion risk, this study found that respondents' education level and geolocation of the households are the strongest predictors of future risk prediction. These findings align with similar research on risk perception. For example, a global study by Lee et al. (2015) found that education level is the single greatest predictor of climate change awareness, while location attributes were found to be one of the greatest predictors. Our analysis also suggests that access to electricity influences the accurate prediction of future erosion risk. Respondents from houses with access to electricity were more accurate in predicting future erosion risk compared to their counterparts. One potential rationale is that an electricity connection represents a type of investment in the property that also increasingly ties such families to the land in a physical sense. This economic investment and hedging on the sedentary nature of their residence may lead families with electricity connections to be more vigilant and aware of potential risks.

Our assessment of perceived and actual risk using a confusion matrix determined that no respondents underestimated their coastal erosion risk. This is in direct contrast to similar studies on risk perceptions in which a direct correlation of households' underestimation of vulnerability to natural disasters was found (Christian et al. 2021; Bonaiuto et al. 2016). This underestimation of coastal erosion risk might be attributed to respondents' previous erosion experience and sociodemographic characteristics (Carlton and Jacobson 2013). Further, underestimation may be influenced by the omnipresent awareness of erosion risk among people living in the region, and it could have been affected by a relatively low number of survey responses (n=214). Thus, we contend that future studies should investigate why residents in the region are not underestimating the risk of erosion—even though this finding could be considered a positive coping mechanism for households and society.

Despite its limitations, this study has provided insights on the accurate prediction of future erosion risk by combing original survey data and coinciding geospatial variables derived by analyzing satellite imagery. Thus, not only were perceptions of erosion risk measured, but actual outcomes of annual erosion were measured to verify or refute actual perceptions of the future. This nuanced approach enables researchers to better assess the accuracy of risk prediction as well as which factors—social, economic, locational—play a role in shaping accurate perceptions. We believe the research findings and approach have potential to advance the knowledge and understanding of coastal erosion risk in Bangladesh and in similarly situated contexts.

5 <u>Conclusion</u>

This study analyzed the drivers of accurate predictions of coastal erosion risk among households living on eastern bank of the lower Meghna River in Bangladesh. We employed both original social survey and satellite imagery data to assess coastal erosion risk. This combination of remote sensing and survey data enabled the investigation of risk perception in an innovative way—literally allowing us to confirm whether surveyed houses were lost to erosion and when. We used a binary logistic regression model to assess what drives accurate prediction of future erosion risk. Among the socio-economic variables, education level ($p = 0.005$) was found to be a significant predictor. Results also determined that proximity to shoreline ($p = <0.001$) and residents without embankments ($p = <0.001$**)** are strongly associated with accurate risk prediction. A majority of respondents believe that a concrete embankment is needed to protect the Meghna shoreline from erosion. Results of this study could be a tipping point for planners, policymakers, non-governmental organizations, and other stakeholders engaged in coastal management in the region. Additionally, this research could help to formulate an integrated coastal development plan in Bangladesh in which embankments and other forms of protective measures may be urgently required to protect the coastal shoreline. This study and its approach could also be used a reference point for similar studies to move towards a better understanding of coastal erosion risk perception.

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Chapter 5. Conclusions

Coastal areas are impacted by erosion globally. The severity and magnitude of coastal erosion is often high in the deltaic environments. Due to erosion, people lose their households, agricultural lands, and livelihoods. Monitoring past and future coastal erosion is of great importance in better planning and management of coastal areas. Using coastal Bangladesh as case study, this dissertation has focused on 1) assessing spatio-temporal empirical forecasting performance of future shoreline positions, 2) evaluating predicted loss of different land use and land cover (LULC) due to coastal erosion, and 3) exploring the patterns and drivers of accurate prediction of coastal erosion risk.

First, this study adopted an innovative data-driven approach to predict future shoreline positions using various statistical methods by examining past shoreline change rates. Using different time depths of shoreline rates, methods of erosion rate calculation, and predicting future shoreline positions for different time horizons, this study assessed the prediction performance of different combinations. This study concludes that the higher the number of shorelines used in calculating and predicting shoreline the better the performance is. It was revealed that the prediction accuracy is generally higher for the near future compared to the distant future. This is a methodological innovation with potential for forecasting applications to other deltas and vulnerable shorelines globally.

Second, this dissertation assessed the impacts of historical and predicted shoreline change on different land use and land covers. A temporal design strategy of three different time depths and four different time horizons to predict future land loss for different scenarios was employed. Results suggest that the best performing model for land loss prediction was 10-year time depth and 20-year time horizon (rank 1) followed by the 5-year time depth and 20-year time horizon model.

Though the results are specific to our study area located in coastal Bangladesh, they can inform and help refine other models to predict land loss due to erosion in other parts of the coastal and deltaic environment worldwide.

Last, this dissertation analyzed the drivers of accurate predictions of coastal erosion risk of the households living in eastern bank of the lower Meghna river region in Bangladesh. Both satellite imagery and social survey data were used. The combination of remote sensing and social survey data enabled to look at risk perception in an innovative way. Using logistic regression, this study found that the physical location of the households like proximity to coast, residents without revetments are strongly associated with the risk prediction. This research will be to formulate an integrated coastal development plan in Bangladesh, in which embankments and other forms of protective measures on the coast are urgently required to protect the coastal shoreline.

Overall, this dissertation advances the scientific understanding of past and future coastal erosion risk and associated changes in land use/cover and coastal risk perception in Bangladesh. While empirical results are specific to the project's study area, results can inform the region's shoreline forecasting ability and associated mitigation and adaptation strategies.